LEVEL 3 | INTEGRATED MONOGRAPH AND COLAB TUTORIAL

THE AUTONOMOUS LEGAL SECOND BRAIN

Endogenous Knowledge Navigation in a Synthetic Mexican Legal System

An integrated academic monograph and sixteen-cell laboratory tutorial on knowledge graphs, autonomous evidence acquisition, comparative evaluation, and committee-style legal memoranda.

CENTRAL PROPOSITION

Autonomy is adaptive evidence selection under explicit constraints, not unrestricted action. A research agent may guide the investigation, but sources, decisions, uncertainty and execution limits must remain inspectable. This laboratory tests that proposition in an entirely synthetic Mexican legal system.

Alejandro Reynoso

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TOPIC 791 | LEVEL 3 LABORATORY
The Second Brain Project | Autonomous Research

The legal corpus, examples and experimental quantities are synthetic teaching constructs, not operative Mexican law or legal advice. Verify current official sources for real legal research.

Abstract

This integrated monograph and laboratory tutorial develops the third laboratory in a progressive curriculum on externalized knowledge systems, or *Second Brains*. Whereas the first laboratory explored semantic retrieval over largely independent notes, and the second introduced traversal of interconnected corporate documents, the third asks whether a language-model-guided agent can decide which legal evidence to investigate next. The setting is an entirely synthetic Mexican legal corpus, deliberately constructed to capture the intellectual features of civil-law research without reproducing or purporting to interpret operative legal provisions. The architecture comprises a 350-document vault, five overlapping relationship graphs, a seed-selection stage, a bounded autonomous research loop, evidence-grounded memorandum generation, and dossier export. Three consultation scenarios and three retrieval strategies permit controlled comparisons of coverage, relevance, citation fidelity, navigation efficiency, and research cost. The central proposition is that autonomy is not synonymous with unrestricted action: meaningful research autonomy requires adaptive selection of evidence under explicit constraints, uncertainty accounting, and auditable decisions.

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1 The pedagogical problem: from retrieval to research judgment

SECTION OVERVIEW

This section distinguishes finding similar text from conducting a defensible research process. It frames the agent as a bounded policy that chooses evidence-seeking actions, records uncertainty, and stops when further investigation is unlikely to change the answer.

A repository of documents is not yet a research system. Its contents may be extensive, carefully categorized, and readily searchable, but the ability to locate a relevant passage differs fundamentally from the ability to determine whether that passage resolves a question. A sophisticated research problem unfolds through successive discoveries. Each discovery may introduce a qualification, expose an exception, identify an institutional competence, or reveal that an apparently peripheral field of law is indispensable to the answer. The researcher must therefore do more than search: the researcher must decide what remains unknown.

The Second Brain sequence isolates this progression. Level 1 treats the vault as a collection of notes and asks whether semantic similarity can surface useful evidence. Level 2 introduces interconnection: ownership links, commercial relationships, and a navigation graph permit evidence to propagate across notes. Level 3 changes the object of instruction. Rather than asking how far a predefined algorithm can travel through a graph, it asks how a research agent can choose the next edge, node, legal domain, or investigative question in response to evidence accumulated so far.

This shift should not be confused with the claim that a language model possesses legal expertise by virtue of producing plausible text. The experiment instead treats the model as a *bounded research policy*: it may select among available tools and articulate provisional judgments, while a deterministic execution layer verifies identifiers, enforces budgets, records provenance, and prevents unsupported claims from silently becoming findings.

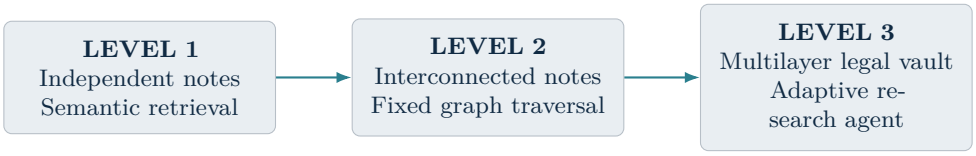


Figure 1: The three-level progression: the crucial change is in who or what chooses the next research action.

Design consideration: Research judgment

The importance of research judgment becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate research judgment into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, research judgment should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Retrieval quality

The importance of retrieval quality becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate retrieval quality into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, retrieval quality should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. The unit of analysis should be a research transition rather than a generated paragraph. At the start of a run, the system records the consultation, its explicit facts, the unresolved predicates, and the initial hypotheses. After each tool call it records what was requested, which sources were returned, whether the result supports or weakens a hypothesis, and what question remains open. That record makes the agent's judgment inspectable: a reviewer can see whether the next step follows from evidence or merely from linguistic association. It also makes negative results useful. An empty search may indicate an unsuitable term, a missing corpus record, or a question that the permitted sources cannot answer; those possibilities should be distinguished before the agent repeats the same search.

The final memorandum should be traceable back through this sequence. Each material claim should identify the supporting synthetic document and state any factual condition on which the claim depends. If the evidence is incomplete, the memorandum should say what is missing instead of filling the gap with a plausible-sounding rule. A bounded system also needs a stopping standard. It may stop when the key issues have evidence, when remaining gaps are explicitly listed, when the action budget is exhausted, or when additional searches repeatedly return redundant material. These are different stopping reasons and should be reported separately. The student can then evaluate not just whether the final answer reads well, but whether the research process was proportionate, reproducible, and appropriately cautious.

Implementation implications

In a working research service, the distinction between a search result and a research decision should be visible in the interface as well as the data model. The intake record should preserve the exact consultation, its date, the requesting person's role, the intended use, and any jurisdiction or time period supplied. A normalized issue statement may be useful for retrieval, but it should remain linked to the unaltered original question so that later reviewers can see whether paraphrasing changed its meaning. Each proposed issue should carry a status such as open, supported, contradicted, fact-dependent, or outside scope. These are workflow states, not conclusions about law.

The action history should make the agent's discretion observable without requiring reviewers to infer it from a final answer. For every action, record the question being investigated, the tool chosen, the constraints applied, the documents returned, and the reason to continue, redirect, or stop. A concise rationale is preferable to an unstructured transcript: it allows a reviewer to evaluate the actual decision and reduces storage of irrelevant prompt material. When the agent changes direction after finding a qualification or exception, the trace should connect that discovery to the new search. When it does not change direction, the trace should still show why the new evidence did not alter the plan.

The system should keep a distinction between evidence collection and synthesis. Retrieval and graph tools return identified records; a separate reasoning stage may compare those records and generate candidate propositions.

A memorandum claim is defensible only if it points to the relevant record and passage, states the proposition supported, and identifies any qualification that limits it. If the supporting record is missing or ambiguous, the system should produce a gap notice rather than fill the space with a plausible completion. This structure also makes correction practical: an editor can amend a source mapping or a fact classification without silently rewriting the entire investigative history.

Stopping should be treated as a research decision with its own record. A run may stop because the action budget is exhausted, because no permitted tool can address the remaining uncertainty, because critical questions have sufficient support, or because a human reviewer must intervene. These outcomes are not equivalent and should not be collapsed into a single success flag. The stopping record should identify unresolved issues, rejected next actions, and the reason the remaining risk is acceptable or requires escalation. In a live workflow, the responsible professional—not the model—must determine whether the evidentiary record is adequate for the intended use. This makes the system useful as an assistant while preserving accountability for the legal work.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Preserve the original consultation alongside every normalized query, issue label, and later revision so reviewers can detect meaning drift.
- Give each decision event a stable identifier and link it to the tool call, returned record IDs, and the next plan state.
- Require every material memorandum proposition to cite a source passage and disclose whether it is direct support, inference, or unresolved.
- Record why the system continued, redirected, escalated, or stopped; never infer a sound stopping decision from a polished final answer.
- Route unresolved legal significance and high-impact outputs to a named human reviewer who records approval, edits, and reasons.

2 Why Mexican legal research is a revealing test case

SECTION OVERVIEW

This section uses Mexican legal research as a synthetic test of source hierarchy, competence, procedure, and interpretation. The example teaches researchers to keep legal priority distinct from relevance while separating invented laboratory materials from operative law.

The Mexican legal setting offers an intellectually rich laboratory because legal consultation frequently requires simultaneous attention to constitutional principles, legislation, implementing rules, institutional competence, administrative procedure, and judicial interpretation. The researcher cannot assume that a semantically similar provision is controlling. Applicability depends on legal hierarchy, the type of transaction, the authority of the source, temporal validity, exceptions, and the facts that have actually been established.

The experiment adopts the broad methodological distinction between civil-law reasoning, in which codified norms and their systematic interpretation are central, and the familiar case-centered pedagogy associated with common-law systems. This distinction is not absolute: judicial precedent and jurisprudence matter in Mexico, and neither legal tradition is reducible to a single research technique. The synthetic environment therefore incorporates jurisprudential interpretation as a distinct graph, without treating all decisions as equally authoritative or all interpretations as mechanically binding.

Constitutional norms supply the normative foundation of the fictional system. They should not, however, become an artificial hub through which every search must pass. An effective agent should identify when constitutional analysis is material and should otherwise prioritize more directly applicable sources. This distinction between *legal priority* and *graph centrality* is one of the most important lessons of the laboratory.

Design consideration: Constitutional supremacy

The importance of constitutional supremacy becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate constitutional supremacy into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, constitutional supremacy should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Competence and jurisdiction

The importance of competence and jurisdiction becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into

one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate competence and jurisdiction into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, competence and jurisdiction should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. The teaching value lies in representing distinctions rather than encoding conclusions about operative law. A record can identify a fictional constitutional principle, a statutory-style rule, a subordinate instrument, an administrative act, or an interpretive decision, while separate metadata describes its issuing body, geographic scope, effective interval, and relation to the hypothetical. The system should never infer legal priority from how often a node appears or how centrally it sits in a graph. A highly connected record may be a convenient starting point without being controlling authority; conversely, a narrowly applicable source may be decisive despite having few edges. The interface and scoring rubric should keep these ideas separate.

The Mexican context also makes competence and procedure visible. A consultation may implicate federal and local institutions, different subject-matter domains, and procedural steps that cannot be collapsed into a single relevance score. The synthetic corpus can represent those distinctions as test conditions: a scenario can vary the location, authority, transaction type, or date and ask whether the agent notices that the research path must change. It should not claim that a fictional record states actual Mexican doctrine. Students should be encouraged to compare the system's classification with official sources during any real-world follow-up, and to treat unresolved jurisdictional or temporal questions as reasons for human review rather than as gaps the language model may fill by inference.

Implementation implications

A real corpus pipeline should be designed as a governed publication process, not as a one-time prompt that generates a folder of plausible documents. Begin with a versioned schema and a data dictionary that defines every field, its accepted values, its provenance, and whether it can be changed by a user, a deterministic validator, or a language model. The document identifier should never depend on a title that may be edited. A separate version identifier should distinguish successive content states, while a stable source key should permit citation continuity where the underlying record is unchanged. Where a source is divided into passages, every passage identifier should retain its parent document and a reproducible locator.

Generation and curation should be separate, logged stages. A generator may produce candidate records, but a validation pipeline should test required fields, dates, document types, synthetic labels, duplicated identifiers, and relationships that point to nonexistent records. Human curators should review both ordinary examples and intentionally difficult cases: superseded texts, partial extracts, similarly named authorities, ambiguous applicability, and conflicting fictional interpretations. Their decisions should be retained as review events rather than written over the generated artifact. When a record is corrected, the old version should remain recoverable for prior experiments, while the new version receives a distinct hash or version stamp.

Corpus balance matters because a retrieval method can appear capable simply because one source type dominates. Domain counts, relation counts, document lengths, exception frequencies, and temporal statuses should therefore be reported before evaluation. The manifest should identify how records were generated, selected, excluded, and reviewed. A set of synthetic records that is internally consistent can still encode unintended assumptions, so review should include a check for whether one fictional domain is systematically treated as more important or more complete than another. Samples should be inspected for wording patterns that leak labels or instructor answers to one strategy more than another.

The production boundary must also be explicit if the fictional corpus is later replaced with genuine legal materials. That transition changes the risk profile: access rights, source freshness, jurisdiction, update responsibilities, retention,

confidentiality, and correction procedures all become operational requirements. Synthetic identifiers and disclaimers should not be reused in a way that makes authentic materials look fictional, nor should fictional text be mixed into production search results. Separate indexes and visibly distinct badges are safer than relying on a hidden metadata flag. Before launch, an owner should be assigned for schema changes, corpus updates, incident response, and removal of records that should no longer be available.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Publish a data dictionary and schema version, and retain the exact manifest used for every research run.
- Store immutable source IDs, revision IDs, content hashes, passage locators, and the curator or process responsible for each change.
- Validate field completeness, temporal ordering, relation endpoints, duplicate records, and synthetic labels before indexing.
- Preserve review decisions and disagreement notes for edge cases instead of silently overwriting generated or sourced material.
- Keep fictional and authoritative production corpora segregated, visibly labeled, and governed by separate update and access policies.

3 The synthetic corpus and its documentary ontology

SECTION OVERVIEW

This section defines the fictional 350-document vault as a structured, reproducible research environment. Its schema distinguishes source identity, legal assertions, factual predicates, temporal status, provenance, and explicit synthetic labeling so retrieval can be tested without implying legal authenticity.

The proposed vault contains 350 compact, fictional documents, generally 100–200 words each. Short documents are an intentional pedagogical choice. The purpose is not to test endurance in long-context reading; it is to isolate the consequences of connectivity, source selection, and research policy. The corpus will be generated reproducibly from a fixed random seed and an explicit schema, with a manifest of identifiers, domains, source types, and relationships.

Each record contains a unique identifier; title; legal domain; source type; short operative text; relevant factual predicates; temporal status; jurisdiction; asserted authority level; cross-references; exceptions; and an explicit `SYNTHETIC` marker. Some records will be deliberately irrelevant despite high semantic similarity. Others will be relevant only if a factual predicate is satisfied. Still others will be superseded, qualified by an exception, or apparently inconsistent with another record. These features create a realistic need for discrimination without pretending to reproduce real Mexican law.

Domain	Documents	Domain	Documents
Constitutional	25	Civil	40
Commercial	45	Criminal	35
Tax	45	Administrative	35
Securities	35	Customs	25
International	30	Jurisprudence	35

Table 1: Proposed distribution of 350 fictional source records.

A small number of highly connected documents will create cross-domain bridges. For example, a fictional securities disclosure provision may connect to corporate representation duties, administrative sanctions, a tax classification rule, and an interpretive decision. Yet not every connection should be equally useful. Relationship labels, applicability conditions, and directionality must be explicit so that traversal does not become indiscriminate association.

Design consideration: Document provenance

The importance of document provenance becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate document provenance into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, document provenance should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Synthetic realism

The importance of synthetic realism becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

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From a methodological perspective, synthetic realism should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. The schema should separate the identity of a document from the propositions expressed in it. Stable identifiers support citations; source type and issuing institution describe provenance; validity intervals and status flags support temporal filtering; factual predicates describe when a fictional rule applies; and typed relations explain why another record is connected. A passage may contain more than one proposition, so a later implementation may store proposition-level identifiers that point back to a document span. This separation prevents a retrieval result from being mistaken for a complete legal conclusion. It also makes it possible to audit which field caused a record to rank highly and to correct a metadata defect without rewriting the source text.

Corpus generation should be controlled by a published schema and a fixed random seed, but reproducibility alone does not guarantee quality. The notebook should validate identifier uniqueness, required fields, allowed source types, date ordering, edge endpoints, and the presence of the synthetic marker before any consultation runs. Human review should inspect a sample from each domain and every deliberately adversarial category, including distractors, superseded records, exceptions, and records whose applicability depends on missing facts. These checks create a known test environment rather than a collection of random legal-sounding sentences. The manifest should record the corpus version and generation settings so two methods are evaluated against exactly the same evidence universe.

Implementation implications

The graph should be treated as a set of claims about relationships, not as a neutral picture of legal reality. Each edge needs a relation type, direction, source or curator, creation time, temporal applicability, and any conditions that limit its use. Where an edge is inferred rather than explicitly stated in a source, the record should say so and identify the inference method. A path through several edges compounds uncertainty: the interface should not display a long chain as if it had the same support as a direct, source-backed relationship. A path summary can show the relation sequence and flag unsupported or low-confidence links before an agent uses it to justify a conclusion.

Relation vocabularies should be developed with domain reviewers and defined in a controlled registry. Terms such as implements, interprets, modifies, distinguishes, requires, may-trigger, and is-subject-to describe different kinds of linkage; they should not be treated as synonyms merely because they connect the same pair of documents. Each definition should include positive examples, exclusions, direction rules, and a known counterexample. Versioning this registry is essential because changing the meaning of a relation can change search behavior even when no document changes. Experimental runs should record the registry version together with the corpus version.

The graph also needs an explicit temporal model. A relationship may apply only during a stated interval, become inactive when a source is superseded, or depend on facts that are not yet known. Queries should be evaluated against a declared as-of date, and the navigation policy should be able to distinguish current, historical, and unresolved edges. For real materials, effective date and publication date may differ; for a teaching corpus, those fields can still

be modeled to teach the distinction. Where two graph paths conflict, the system should present both paths and the reason for the conflict rather than selecting the path with the most edges or the highest numerical score by default.

Graph quality should be evaluated separately from agent quality. A missing bridge can prevent any method from finding a relevant document, while an erroneous edge can steer every method toward a false association. Maintain a graph audit view that lets a reviewer inspect high-degree nodes, orphan records, relation types with unusual frequencies, and edges that have never been reviewed. Test navigation after removing or perturbing selected links. If a method's performance depends on one manually curated edge, that dependence should be reported. This analysis makes it possible to improve the graph without attributing every retrieval failure to the agent or every success to autonomous reasoning.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Require every edge to carry a controlled relation type, direction, source span, applicable dates, and review status.
- Show the full typed path behind each recommendation, including inferred links and uncertainty accumulated across the path.
- Version the relation vocabulary and graph snapshot so results can be reproduced after links or definitions change.
- Provide reviewer tools for orphan nodes, unusual hubs, stale links, conflicting paths, and unverified cross-domain bridges.
- Test whether conclusions survive removal or alteration of key edges, and disclose graph weaknesses that materially affect the result.

4 The five-layer legal knowledge graph

SECTION OVERVIEW

This section models one shared document vault through five typed relationship layers: normative hierarchy, substantive obligations, jurisprudence, procedure and jurisdiction, and cross-border regulation. The graph must preserve relation meaning and provenance rather than treat every connected document as equally authoritative.

The vault is one collection of documents with five overlapping graphs, not five isolated repositories. Formally, let V denote the document set and let E_k denote the typed edges in layer k. The legal knowledge structure is

$$\mathcal{G} = (V, E_1, E_2, E_3, E_4, E_5, \mathcal{A}), \tag{1}$$

where $\mathcal{A}$ contains document and edge attributes, including source type, applicability conditions, and validity status.

The *normative hierarchy graph* records implementation, delegation, qualification, and priority relations. The *substantive relationship graph* connects obligations and rights across contractual, corporate, tax, criminal, and property questions. The *jurisprudence graph* records interpretations, distinctions, exceptions, and possible tensions among fictional decisions. The *procedure and jurisdiction graph* links authorities, procedural acts, remedies, deadlines, and evidentiary obligations. The *cross-border and regulatory graph* relates international transactions to customs, securities, investment, and other regulatory concerns.

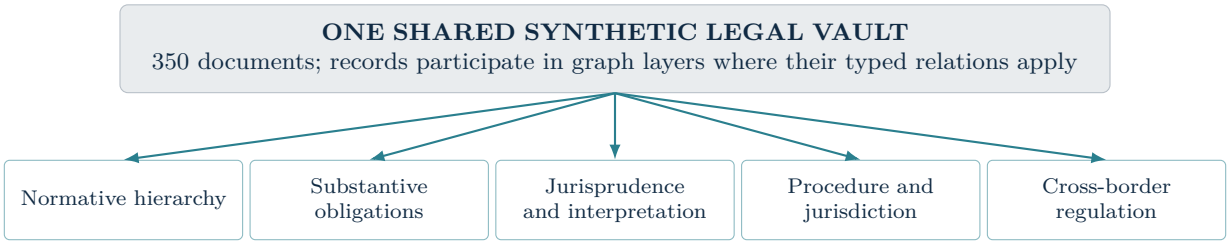


Figure 2: Five typed, overlapping relationship graphs organize one shared corpus; the layout does not encode a hierarchy among the graph layers.

A cross-layer bridge is meaningful only when its semantics are documented. An edge labeled `implements` is not interchangeable with one labeled `may-trigger`, and an interpretive relationship is not equivalent to a procedural prerequisite. The agent's tool interface should expose these labels and their conditions rather than merely returning undifferentiated neighbors.

Design consideration: Typed relations

The importance of typed relations becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate typed relations into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, typed relations should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions

are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Graph directionality

The importance of graph directionality becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate graph directionality into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, graph directionality should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. The layers are overlapping views over shared records, not five separate collections and not five ranks in a hierarchy. One fictional document may participate in a normative edge because it implements another record, a substantive edge because it conditions a contractual obligation, and a procedural edge because it triggers a filing step. Each edge therefore needs a type, direction, source, confidence or review status, and applicability condition. A traversal engine should not treat an edge labeled `interprets` as equivalent to one labeled `implements`; nor should it assume that relationships are transitive. If record A implements B and record B refers to C, it does not follow automatically that A governs C.

Graph diagnostics should test both structure and meaning. Structural checks can find orphan records, duplicate edges, invalid identifiers, and unintended disconnected components. Semantic review should inspect a sample of each edge type and test whether its direction matches the stated relation. The evaluation can then ask whether a relevant cross-layer path was reachable and whether its intermediate links were justified. Centrality and degree may help identify navigational hubs, but they are not substitutes for authority, scope, or applicability. A graph visualization should make its levels legible without implying a false ranking; this is why the corrected diagram centers each row under the shared constitutional frame and labels the vault as a common substrate. The diagram is an explanatory map, not itself a legal ontology.

Implementation implications

Autonomous navigation becomes inspectable when each cycle has a stable input, a limited choice set, an observable result, and a stated reason for the next move. Before the first action, the agent should receive a compact research state: the consultation, the current issue list, known and missing facts, sources already considered, remaining resource budget, and permitted tools. After a tool returns, the state should be updated from verified data rather than from an unvalidated model summary. This separation prevents the model from becoming the sole authority on what has already been found and reduces accidental repetition or loss of a critical qualification.

The action protocol should distinguish searching for a document, reading a passage, following a typed relation, checking temporal status, asking a clarification question, and recommending human escalation. Each action should have a machine-readable schema, maximum result size, timeout, and failure behavior. Invalid actions should be rejected by the executor and recorded; the model should then receive a constrained explanation of the error and a chance to select an allowed alternative. The executor should never turn free-form model text into a database query or external side effect without validation. In a production setting, read-only research actions should be separated from actions that could contact a person, file a document, change a record, or make a decision.

A useful policy can rank candidate actions by expected reduction in a defined research gap, but a score is not self-justifying. The system should expose the signals used: issue relevance, source authority, time fit, novelty, likely exception value, expected cost, and uncertainty. The score should be treated as a prioritization aid, not a legal conclusion. A simple explanation can state which missing proposition the action is intended to test and what result would change the plan. This creates a testable link between stated rationale and observed outcome. If the retrieved material does not answer the question, the next step should reflect that failure rather than quietly treating a tool call as progress.

Budgets should be multidimensional. A maximum number of steps alone may reward many tiny calls or permit one oversized retrieval. Set limits on calls, documents inspected, context tokens, elapsed time, model retries, and external service cost. Each counter should be visible to the agent and logged by the executor. Reserve an explicit intervention route for a human to stop, narrow, or redirect the task. A bounded system should fail safely when a source is unavailable, a result is malformed, or the remaining budget cannot support a defensible analysis. It should report the constraint and unresolved issue instead of manufacturing completeness.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Log every navigation cycle as state-in, proposed action, validated action, tool result, updated state, and reason for the next step.
- Enforce machine-checked schemas, allowlists, timeouts, and independent executor validation for all agent-requested tools.
- Show why each selected action addresses a named gap and which expected result would change the research plan.
- Track separate limits for calls, documents, tokens, elapsed time, retries, and monetary cost; log each limit event.
- Provide a human stop or escalation control and return explicit unresolved issues when tools fail or budgets expire.

5 From Cell 7 to Cell 8: endogenous navigation

SECTION OVERVIEW

This section explains the transition from seed selection to adaptive evidence gathering. The agent proposes a next action from a bounded tool set; a deterministic executor validates it, records its result, enforces budgets, updates open questions, and applies explicit stopping rules.

Cell 7 retains a useful function from the previous laboratory: it transforms the consultation into initial issues and semantic seeds. Its output is not a conclusion but an agenda for investigation. The new Cell 8 becomes an iterative policy that receives the consultation, current evidence, unresolved issues, candidate navigation actions, and remaining budget. It chooses whether to inspect a document, traverse a typed relation, search within another domain, test an exception, or stop.

Let q be the consultation, E_t the collected evidence, U_t the unresolved issue set, B_t the remaining budget, and a_t the next action. The agent policy is represented as

$$a_t = \pi_{\text{LLM}}(q, E_t, \mathcal{G}, U_t, B_t). \tag{2}$$

The research state evolves after tool execution:

$$(E_{t+1}, U_{t+1}, B_{t+1}) = F(E_t, U_t, B_t, a_t, o_t), \tag{3}$$

where o_t is the validated tool observation. This representation does not imply that the model sees the entire graph. The tool gateway exposes only bounded, relevant candidate nodes, metadata, and neighborhood summaries.

The distinction between endogenous and exogenous control is fundamental. The LLM may choose the next domain, document, and line of inquiry. The runtime imposes a hard ceiling on steps, retrieved records, tokens, and spending; validates node identifiers; detects repeated or unproductive cycles; and requires a reasoned stop decision. Reaching a ceiling is a budget exhaustion event, not proof that the legal research is complete.

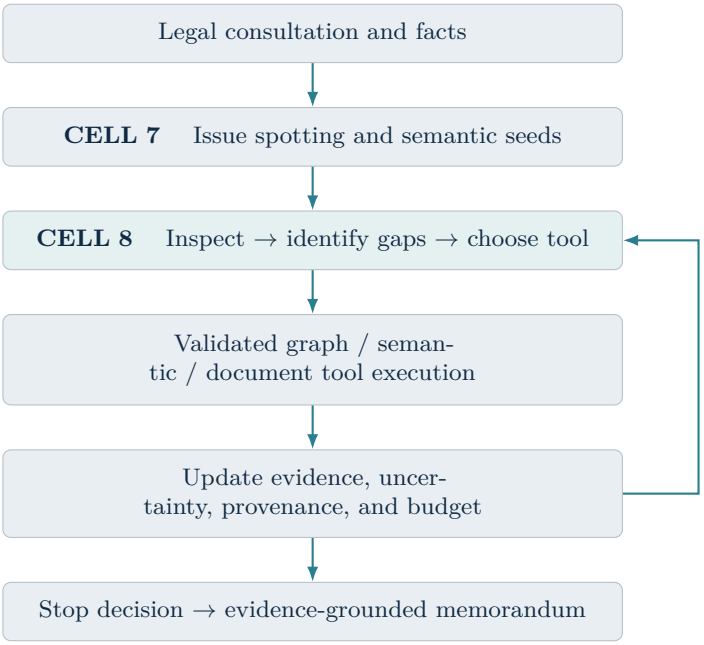


Figure 3: The autonomous research loop: adaptive selection of evidence under deterministic execution safeguards.

Design consideration: Decision policy

The importance of decision policy becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory

instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate decision policy into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, decision policy should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Tool execution

The importance of tool execution becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate tool execution into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, tool execution should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. The agent's action set should be deliberately small and typed. It may search by phrase or metadata, inspect a document, retrieve a node's typed neighbors, test a relation, or request a bounded summary of the evidence state. Each action has an explicit input schema and a deterministic handler. The model can propose an action and explain its purpose, but the executor checks that the target identifier exists, that the operation is allowed, and that the remaining budget is sufficient. The system stores the raw tool result separately from the model's interpretation of it. This separation makes it possible to examine whether an error arose from retrieval, data structure, reasoning, or final synthesis rather than treating the whole loop as an opaque answer generator.

Budgets should cover more than a single maximum number of iterations. A run can have limits on documents read, total tool calls, model tokens, elapsed time, and repeated queries. Reaching one limit should create a recorded stop reason and a concise account of unresolved issues. The policy should also be able to stop early when it has adequate evidence, because exhausting the budget is not a research objective. Conversely, the agent should not terminate merely because it has found one highly relevant document if the consultation signals an exception, a separate competence question, or a material missing fact. This distinction makes bounded autonomy adaptive but not open-ended: the loop can revise its priorities while the tool interface, budgets, provenance records, and stop conditions remain fixed and inspectable.

Implementation implications

An operational consultation should be represented as a structured, revisable fact matrix rather than a single compressed narrative. Each fact needs a text span in the original consultation, a normalized statement, a status, a date or relevant time interval, and a note identifying what evidence could verify it. The status vocabulary should distinguish established within the hypothetical, reported but unverified, disputed, absent, and unknown. These labels should not be changed merely because a retrieved document makes one version of the facts seem convenient. When new information arrives, the previous state should remain visible and the update should record who supplied it and when.

Issue spotting can then be modeled as conditional branching. For each candidate issue, record the factual predicates that would make it relevant, which predicates are satisfied, which remain unknown, and what research or clarification could resolve them. The system can prioritize an issue without asserting that the legal condition has been met. For example, a cross-border financing question may warrant investigation while the investor's legal status or transaction instrument remains uncertain. The output should state how the analysis changes under plausible alternatives rather than hide those alternatives inside a single confident answer. This approach makes the memorandum useful for planning and due diligence while keeping fact uncertainty distinct from uncertainty about the fictional rule.

The final memorandum should have a stable structure that resists unsupported synthesis: question presented, facts and their status, issues, evidence reviewed, conditional analysis, contrary or limiting material, missing information, and proposed next steps. Each legal proposition should have a source citation that resolves to the exact synthetic record and passage. The writer should differentiate quotations, paraphrases, inferences, assumptions, and recommendations. A claim may be supported by several documents, but the memorandum should explain their distinct roles instead of appending a long list of references. Where sources disagree, both should be identified and the disagreement should be described in neutral language.

Consultation variants can help test whether the system responds to facts rather than wording alone. Create paired cases that differ in one material fact, then compare the issues selected, evidence searched, and conclusions qualified. A robust workflow should change course when the material fact changes, while remaining stable under irrelevant edits such as a harmless change in names or sentence order. This is a practical diagnostic for shallow keyword matching. It also teaches that a research memorandum is conditional: a fact change can alter applicability, but the system must show which premise caused the change and must not imply that a fictional outcome predicts a real legal result.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Preserve each source fact, normalized statement, status, time reference, and verification trail as separate linked fields.
- Keep prior fact states and record who changed a classification, when it changed, and what evidence justified the update.
- Link each issue to its factual predicates and show which unknown or disputed fact prevents a firm conclusion.
- Require memorandum claims to distinguish source-supported statements, inferences, assumptions, and recommendations with pinpoint citations.
- Test paired consultations that vary one material fact and retain the trace explaining why the research path or conclusion changed.

6 The synthetic consultation and legal reasoning

SECTION OVERVIEW

This section turns three fictional consultations into structured research tests. Each scenario distinguishes established facts from missing predicates, invites issue spotting without assuming liability, and requires the memorandum to connect conditional conclusions to retrieved sources and unresolved questions.

The primary consultation concerns *Tecnología del Pacífico, S.A. de C.V.*, a fictional Mexican technology company that received foreign financing, imported equipment, and entered several commercial agreements. Management may have made misleading investor representations, misclassified imports, and failed to satisfy tax obligations. An investor threatens litigation, an authority requests information, and the board seeks an assessment of legal exposure and immediate steps.

This consultation is intentionally underdetermined. The agent should not infer criminal liability from suspicion, assume a securities offering occurred merely because investors are involved, or assert that every cross-border payment triggers a particular tax result. Instead, it should distinguish known facts from missing predicates and navigate accordingly. A strong response identifies possible legal domains, tests applicability, documents uncertainty, and specifies which additional facts or sources would be necessary to resolve the remaining questions.

Two further consultations test transfer within the same vault: a customs investigation and a cross-border corporate acquisition. These scenarios should produce different paths, not merely different final prose. The pedagogical objective is to expose how the same document network supports multiple issue-specific research policies.

Design consideration: Fact qualification

The importance of fact qualification becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate fact qualification into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, fact qualification should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Legal issue spotting

The importance of legal issue spotting becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate legal issue spotting into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, legal issue spotting should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. The consultation should first be converted into a fact matrix with four labels: established within the hypothetical, reported but unverified, expressly absent, and unknown. For the technology company, foreign financing is stated, while the precise instrument, investor status, representation text, import classification, relevant dates, and authority request may need qualification. The agent should not silently promote an allegation into an established fact. Instead, it can identify which legal issue depends on that fact and either search for a source that defines the predicate or ask a targeted follow-up question. This method makes issue spotting conditional and supports a memorandum that distinguishes likely pathways from conclusions that cannot yet be reached.

The three consultations can be made comparable without making them interchangeable. The customs scenario should emphasize classification, documentation, shipment facts, and administrative process; the acquisition scenario should emphasize corporate authority, transaction structure, cross-border review, and sequencing of diligence; the primary scenario should require the system to notice interactions among financing, disclosure, imports, tax, and possible disputes. A controlled variant changes one material fact at a time, such as the date, jurisdiction, identity of the investor, or status of an agency request. Reviewers can then see whether the research path changes for a reason connected to that variation. The final memorandum should use a stable structure—question presented, established facts, issues, evidence, conditional analysis, missing facts, and next steps—while making clear that it is an educational output grounded only in fictional materials.

Implementation implications

A defensible comparison begins by defining the unit of analysis. A run may be one method applied to one consultation under one corpus and configuration; a claim-level assessment may be one proposition in one final memorandum; a navigation measure may be one validated tool action. These units should not be mixed. For each measure, state the denominator, the treatment of missing observations, and whether higher or lower values are preferable. Evidence recall, for example, should specify the critical-record set and whether partial retrieval counts. Citation fidelity should distinguish a correct document from support for the exact proposition. A plausible source name with no relevant passage is not a supported citation.

Methods should receive equivalent opportunities but not be artificially forced into identical mechanics. Semantic search, deterministic traversal, and adaptive navigation differ in their natural action spaces. Fairness means controlling the evidence universe, consultation wording, time horizon, output format, and practical budget, then transparently reporting any remaining difference. Where autonomous research uses a language model but a baseline does not, a separate answer-generation control can help separate retrieval gains from prose-generation effects. A secondary analysis can compare methods at matched resource levels or plot quality against cost instead of reducing every outcome to one arbitrary score.

A single run is rarely enough to characterize a stochastic system. Repeat evaluations across declared seeds, prompt variants, and consultation variants, while preserving paired conditions so each method faces the same case. Report distributions and uncertainty, not only a best score or a single average. Keep a record of failed or incomplete runs and define in advance whether they count as operational failures, exclusions, or missing data. If a service error prevents a method from finishing, that event is relevant to deployment reliability even if it cannot be scored for legal accuracy. A prewritten analysis plan makes it harder to choose only the runs that support a preferred conclusion.

Human assessment should use blinded materials where practical. Assessors can review claim-source pairs without knowing which method produced them, apply a rubric with examples, and separately rate support, qualification, citation precision, and treatment of uncertainty. Disagreements should be preserved and resolved using a defined

procedure; inter-reviewer agreement is informative and should not be hidden behind consensus alone. The instructor key should be locked before outcomes are examined, with revisions versioned and justified. These practices do not create perfect ground truth, but they make the limits of the ground truth inspectable and allow readers to distinguish an agent's performance from the assumptions embedded in the scoring instrument.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Define each metric's unit, denominator, scoring rule, missing-data policy, and interpretation before examining the comparative results.
- Preserve paired run IDs, consultation versions, corpus versions, seeds, prompts, model settings, resource use, and failure status.
- Blind claim-support reviewers to method identity where feasible and retain individual scores, rationales, and disagreement records.
- Compare methods at transparent resource levels and report quality-cost tradeoffs rather than relying on one opaque aggregate score.
- Freeze and version the instructor key and analysis plan before scoring; document every later amendment and its rationale.

7 The experimental comparison

SECTION OVERVIEW

This section compares semantic retrieval, fixed graph traversal, and bounded autonomous navigation on common synthetic consultations. A shared scoring key evaluates evidence coverage, citation fidelity, unsupported claims, navigation efficiency, research cost, and stopping quality.

Three methods should operate on the same corpus and consultation set. Method A performs semantic retrieval only. Method B uses deterministic traversal with predefined graph relations and hop limits. Method C uses the LLM-guided research policy with bounded tool access. All methods should use a common final-answer template and, as far as practicable, comparable evidence and model budgets, so that gains are not merely consequences of allowing one method to read much more material.

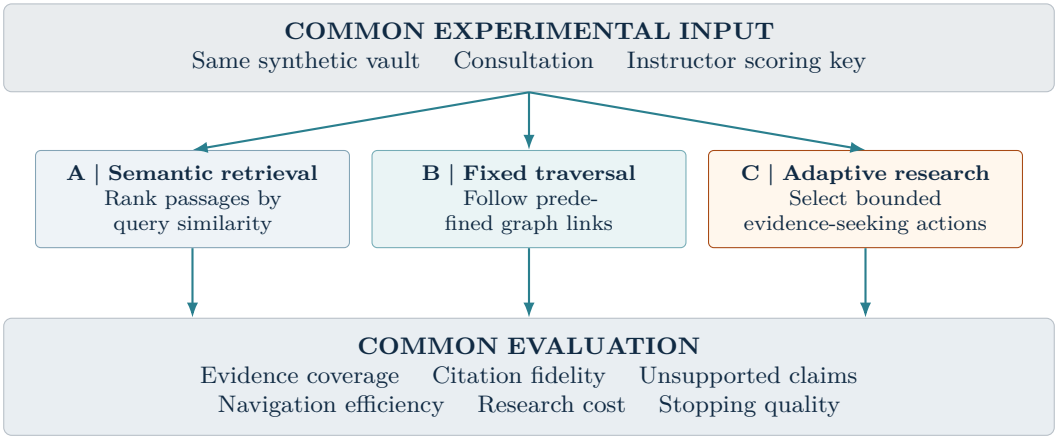


Figure 4: Three research strategies receive the same inputs and are assessed against a common evidence and process rubric.

A hidden instructor key will specify critical synthetic authorities, necessary cross-domain bridges, distractors, exceptions, and unresolved factual predicates. Evidence recall measures the proportion of critical authorities found. Citation fidelity assesses whether claims point to appropriate documents. Unsupported-claim rate penalizes conclusions not justified by retrieved evidence. Navigation efficiency relates useful evidence to tool actions or cost. Stop quality tests whether the agent halts with important questions unresolved or continues after diminishing returns.

The experiment must be capable of disproving its motivating intuition. An autonomous method that makes more calls but fails to improve coverage or citation quality has not demonstrated superiority. Likewise, a deterministic baseline may perform well when graph structure is simple and complete. The pedagogical value lies in the comparative evidence, not in a predetermined victory for autonomy.

Design consideration: Baseline comparability

The importance of baseline comparability becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate baseline comparability into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, baseline comparability should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Ground-truth construction

The importance of ground-truth construction becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate ground-truth construction into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, ground-truth construction should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. Fair comparison requires an identical evidence environment, not merely similarly named methods. All three strategies should receive the same consultation wording, corpus version, metadata, permitted source universe, and final response template. If the autonomous agent receives a larger evidence or token budget, that difference should be recorded and analyzed rather than hidden. Repeated runs should use paired scenario instances and preserve random seeds where applicable. A deterministic baseline should be run first as a reproducibility check; model-based runs can then be compared under the same declared settings. These controls do not remove all variability, but they make its sources visible and limit the temptation to attribute every difference to the navigation policy.

The metrics should be operationalized before results are inspected. Evidence recall is the share of instructor-designated critical records retrieved; precision can reveal whether broad retrieval overwhelms the answer with distractors. Citation fidelity should separately measure identifier correctness and whether the cited record supports the attached proposition. Unsupported-claim rate can be assessed claim by claim using a rubric that distinguishes direct support, qualified inference, and no support. Efficiency may be reported as relevant evidence per call and total cost, but should not reward a short run that misses necessary issues. Stop quality can be scored against known unresolved predicates and diminishing returns. Results should include per-scenario values and variation across runs, not only one aggregate ranking. The scoring key and adjudication process should be documented so that the experiment can reveal a null or negative result as readily as an apparent advantage.

Implementation implications

The notebook should behave like a small research application whose state is explicit, rather than a sequence of cells that only works in the author's current runtime. Each cell should state its purpose, inputs, outputs, and expected side effects. Setup should be idempotent where possible: rerunning it should not duplicate records, silently change corpus state, or retain a previous consultation. A reset function should return the notebook to a known baseline. Before learners run the agent cell, they should be able to inspect the data manifest, graph summary, tool schemas, action budget, and a sample of records. This makes the experiment teachable before any model call is made.

The ten-cell structure should reveal the dependency chain from initialization through report generation. Separate

configuration from execution, data generation from validation, retrieval from reasoning, and scoring from presentation. If one cell fails, the notebook should identify the affected artifact and tell the learner what state remains valid. Model calls should be optional or gated by an explicit configuration flag so deterministic demonstrations remain possible when credentials, connectivity, or budget are unavailable. A fallback should be clearly labeled as deterministic; it should never impersonate an autonomous result for the sake of a smooth demonstration.

Reproducible execution requires a run manifest saved with outputs. At minimum, it should include the notebook revision, code version, runtime date, corpus and graph versions, consultation identifier, random seed, model and prompt configuration, tool limits, and evaluation version. Sensitive keys must be read from a secret manager or environment setting, never embedded in a cell or printed in diagnostic output. The notebook should avoid storing personal consultations in shared artifacts by default. A clean-runtime test should start from a fresh session, run cells in order, reset and run again, and compare deterministic outputs. Any expected nondeterminism should be described rather than treated as a defect.

The learner-facing report should explain what the run establishes and what it does not. It can display retrieved evidence, typed paths, action trace, remaining uncertainty, metric definitions, and costs. It should not present the synthetic memo as advice or imply that an experimental score validates real legal research. Exported artifacts need a synthetic-data banner and should retain provenance if downloaded or copied. Notebook instructions should also specify how to report a defect without sharing API secrets or sensitive case facts. These modest operational practices transform a code demonstration into a reproducible, inspectable exercise that instructors can safely adapt.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Put each cell's purpose, required inputs, outputs, side effects, and recovery procedure in a short, visible description.
- Save a run manifest with notebook revision, corpus snapshot, consultation ID, seed, model settings, prompts, and budget configuration.
- Test clean-runtime execution and reset behavior; record nondeterministic outputs and any cell that depends on hidden state.
- Store credentials outside notebook text and outputs, and prevent confidential consultations from entering shared logs or exports.
- Label deterministic fallback runs and synthetic outputs explicitly, and preserve evidence IDs and metric definitions in exported reports.

8 The original ten-cell laboratory blueprint

SECTION OVERVIEW

This section maps the experiment into ten executable notebook cells, from environment setup and synthetic data generation through adaptive research, comparison, reporting, and interface. Each cell has an observable output and can be checked without hiding failures in an end-to-end run.

The implementation is designed as exactly ten executable code cells, each preceded by an approximately 300-word explanation. A roughly 1,000-word introduction motivates the experiment and a roughly 750-word conclusion interprets results, limitations, and implications for institutional knowledge systems. The execution environment is Google Colab, with a model gateway using Claude Haiku 4.5 when available and explicit deterministic fallbacks for failures.

Cell	Responsibility	Observable output
01	Environment and model gateway	Dependency checks, secret loading, model availability, robust response parsing.
02	Synthetic corpus generation	Reproducible vault of 350 records and manifest.
03	Multilayer graph construction	Typed edges, integrity tests, network summaries.
04	Retrieval and graph tools	Semantic search, neighbor inspection, document reading, cross-layer lookup.
05	Consultations and gold keys	Three fact patterns, critical source sets, known distractors and missing facts.
06	Baseline methods	Semantic-only and deterministic navigation traces.
07	LLM seed selection	Issue list, initial evidence candidates, explicit uncertainties.
08	Autonomous research loop	Auditable tool decisions, state updates, adaptive paths and stopping reason.
09	Grounded memorandum	Issue-by-issue analysis with source IDs, assumptions and open questions.
10	Visualization and evaluation	Path diagrams, domain coverage, quality and cost comparison.

Cell 8 is the conceptual center but should not absorb the entire implementation. Its tools, graph representation, and evaluation criteria are prepared in earlier cells. Each decision should be logged with iteration number, selected action, target node, stated research objective, tool result, budget use, and any change in unresolved issues. This makes the exercise inspectable by students who do not need to understand the underlying model implementation in detail.

Design consideration: Ten-cell pedagogy

The importance of ten-cell pedagogy becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate ten-cell pedagogy into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, ten-cell pedagogy should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than

sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Modular execution

The importance of modular execution becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate modular execution into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, modular execution should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. The notebook should behave like a sequence of inspectable experiments rather than a monolithic script. Each cell should declare its inputs, create named outputs, and report checks that a learner can verify before proceeding. Corpus generation should expose the seed and manifest; graph construction should show node and edge counts with integrity tests; tool cells should demonstrate each allowed operation on known records; and consultation cells should preserve the fact pattern and instructor key separately. The autonomous loop should emit a compact trace after each iteration, while the comparison cell should run all methods under the same scenario configuration. The report cell should expose both metrics and examples of supported, unsupported, and unresolved claims.

Execution should remain possible when the model service is unavailable. A deterministic fallback can demonstrate data flow and baseline evaluation, but its output must be labeled as such and must not be presented as an autonomous run. API credentials should be read from Colab Secrets at runtime and never placed in notebook text, outputs, or saved logs. Cells should fail with actionable messages when a required secret or dependency is absent, and downstream cells should not continue with empty or stale state. A reset mechanism should restore the same corpus and scenario conditions before paired comparisons. These choices make the notebook useful both for teaching and for review: students can rerun one stage, inspect intermediate artifacts, and locate defects without losing the provenance of the overall experiment.

Implementation implications

A safeguard program should begin with a threat model that names assets, actors, failure paths, and consequences. Relevant assets include source integrity, confidential consultations, credentials, evaluation keys, and the audit record. Threats include prompt injection embedded in a document, a model inventing a source, an executor accepting an invalid identifier, accidental disclosure through logs, unauthorized access to a sensitive matter, and an operator editing a scoring key after observing outcomes. Controls should be attached to these concrete risks and tested, rather than described as general commitments to responsible AI.

The corpus should be treated as untrusted input even when the exercise is synthetic. Retrieved text may contain instructions that attempt to redirect the model, disclose data, or bypass tool constraints. The model should receive clear separation between system instructions and source excerpts, while the executor independently checks every requested action. The model must not be able to execute arbitrary code, browse unrelated sites, or change corpus

records. Output validation should detect malformed citations, nonexistent IDs, unsupported quotes, and claims that lack a source mapping. If validation fails, the system should withhold or qualify the output and identify the defect for review.

Scientific validity is also a safeguard against misleading institutional confidence. A strong score on a small, synthetic corpus does not establish legal competence, and a low unsupported-claim rate may simply reflect a cautious output format. Construct validity asks whether the metrics represent the desired capabilities; internal validity asks whether method differences, rather than budgets or data exposure, explain observed differences; external validity asks whether findings transfer beyond the designed scenario. Each must be discussed separately. Robustness checks should vary corpus balance, source ambiguity, fact completeness, prompt wording, and graph density, then report whether a result is stable across these changes.

Governance should specify who may run the system, what information may be entered, how long traces are retained, who can inspect them, how incidents are escalated, and how an affected user can request correction. Logs should be sufficient to reconstruct decisions but should not become an uncontrolled copy of confidential case material. Retention schedules, access controls, redaction, and deletion procedures must therefore be designed together. For real legal workflows, review by information security, privacy, professional responsibility, and subject-matter specialists should precede any pilot. An educational prototype can teach these requirements, but it cannot waive them or establish that a deployment is safe merely because the underlying examples are fictional.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Maintain a documented threat model for source integrity, confidential inputs, credentials, scoring keys, and model-generated actions.
- Test prompt-injection, invalid-citation, malformed-output, unauthorized-tool, and data-leakage cases before releasing a workflow.
- Separate evidence of construct, internal, and external validity; report robustness limits instead of generalizing from one synthetic score.
- Define role-based access, redaction, retention, deletion, correction, and incident-escalation rules for every stored trace.
- Require security, privacy, professional-responsibility, and subject-matter review before any pilot with real legal materials or matters.

9 Operational safeguards and scientific validity

SECTION OVERVIEW

This section establishes boundaries for source authenticity, credentials, model behavior, oversight, and reproducibility. Synthetic content must remain unmistakably fictional, evaluation claims must be falsifiable, and consequential legal judgments must remain outside the agent's authority.

The corpus is fictional by design. Every document and simulated decision must carry an unmistakable synthetic label, and the generated memorandum must not be represented as Mexican legal advice. The fictional rules should be internally coherent and conceptually inspired by civil-law institutional structure, but they should not impersonate operative statutory text or real Supreme Court jurisprudence. Real-world validation would require authoritative sources, legal currency checks, and professional review beyond the scope of this laboratory.

The Colab should obtain the API credential from `ANTHROPIC_API_KEY` in Colab Secrets, never print the key, and permit a reproducible non-LLM demonstration if the gateway is unavailable. The model adapter should avoid assuming that every model accepts a `temperature` argument, should validate structured responses, and should recover from malformed JSON, timeouts, and length limits. When the fallback is used, plots and reports must label it honestly as deterministic rather than autonomous.

Evaluation should be repeated across scenarios and, when resources permit, across several model runs. A single attractive navigation diagram is not evidence of robust performance. Results should preserve run identifiers, corpus version, random seed, model identifier, prompt version, tool-call logs, and evaluation settings. The aim is to make autonomy observable and falsifiable.

Design consideration: Source authenticity

The importance of source authenticity becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate source authenticity into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, source authenticity should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Human oversight

The importance of human oversight becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented

provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate human oversight into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, human oversight should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. Security and integrity controls should be treated as part of the research method. The notebook must not hard-code secrets, print credentials, or include sensitive user information in model prompts or logs. Although the corpus is fictional, prompts and outputs should still be inspected for accidental disclosure and for text that could be mistaken for genuine legal authority. A synthetic marker should travel with every record into search results, graph views, citations, and generated memoranda; relying on a single introductory disclaimer is insufficient if a later excerpt is copied out of context. The model should also be treated as an untrusted component: validate its requested tool name and arguments, reject unknown identifiers, and render tool results from the corpus rather than allowing the model to invent source content.

Scientific validity requires attention to what each score actually measures. A retrieval metric can demonstrate that a known synthetic item was found, but it cannot establish that the item would be legally controlling in practice. The instructor key itself may encode assumptions, so its construction should be documented, reviewed independently, and versioned with the corpus. More than one reviewer can assess a sample of claims and disagreements should be reported rather than erased by averaging. Robustness checks should vary query phrasing, missing facts, distractor density, and graph connectivity to identify when the method fails. All runs should preserve prompts, model identifier, corpus version, settings, traces, and scoring decisions, while avoiding secrets. The central safeguard is proportionality: the experiment may teach evidence navigation and auditability, but any use involving real people, operative sources, or legal decisions requires a separate authority, privacy, and professional-review framework.

Implementation implications

Results should be interpreted at three distinct levels. The first is operational: did the system complete the task within its limits, preserve the trace, and identify its failures? The second is evidentiary: did it find relevant records, connect them through supportable relationships, and represent uncertainty accurately? The third is institutional: did the resulting work product help a responsible person make a better-informed decision without obscuring who is accountable? Success at one level does not prove success at another. A system may be operationally reliable but retrieve weak evidence; it may retrieve good evidence yet present it in a misleading form; or it may be technically strong but unsuitable for a particular organization's authority structure.

An institution adopting this architecture should begin with a narrow, low-consequence use case and a defined supervisor. That supervisor should be able to see the original question, evidence trail, unresolved issues, and limits, and should have authority to reject the output. Workflows should identify what can be delegated to the agent, what requires a professional judgment, and what is prohibited. The right boundary depends on consequence, reversibility, confidentiality, and the availability of qualified review. A research agent that only proposes source passages is materially different from one that drafts a client-facing opinion or initiates a procedural action. Those capabilities should not be grouped under a broad label of autonomy.

Adoption should also be evaluated through organizational costs and incentives. Faster production is not necessarily better if reviewers spend more time detecting unsupported assertions or if users over-trust fluent summaries. Measure the time required to verify a claim, correct a source mapping, identify a missing issue, and reproduce a result. Collect feedback from researchers, supervisors, and affected teams, not only from system builders. Review whether the interface makes uncertainty and contrary evidence visible or whether it rewards the appearance of decisiveness. The model's behavior is shaped by the task, evaluation, interface, and institutional incentives; governance must address the whole service rather than only the model weights.

Transfer to other domains should be framed as a hypothesis. Financial due diligence, compliance review, and

regulatory analysis may share needs for cross-domain evidence and traceable action selection, but their source authority, risk tolerance, and decision rights differ. Before transfer, map the new domain's authoritative materials, relevant facts, temporal rules, exception types, and escalation pathways. Then rebuild the test set and scoring rubric with domain experts. The legal experiment may contribute design principles such as typed provenance and bounded navigation, but it does not authorize reuse of its synthetic ontology or demonstrate that performance will transfer unchanged. Each institution should be able to explain why the system is appropriate for its own tasks and how a reviewer can challenge its use.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Separate operational reliability, evidentiary quality, and institutional suitability in deployment reports and approval decisions.
- Name an accountable supervisor, define permitted and prohibited uses, and give that reviewer practical authority to override or halt.
- Measure verification effort, correction burden, issue omissions, user over-reliance, and adverse effects alongside speed or answer quality.
- Revalidate the source map, fact model, review roles, and score rubric with domain experts before transferring the design to another field.
- Preserve a contestable record that lets affected reviewers inspect evidence, challenge assumptions, and request a corrected analysis.

10 Interpretation: knowledge, navigation, and institutional judgment

SECTION OVERVIEW

This section interprets performance as evidence about a research policy, not proof of legal expertise. It asks which paths produced useful discoveries, where uncertainty or provenance failed, how navigation costs changed, and which conclusions should remain provisional.

A Second Brain becomes more useful when it can connect evidence across domains. It becomes qualitatively different when it can choose its own research trajectory in response to discovered gaps. This does not mean that the agent acquires legal authority, professional responsibility, or an unrestricted mandate. The appropriate analogy is a supervised research assistant with a structured corpus, constrained tools, an explicit audit trail, and a duty to distinguish evidence from inference.

The legal laboratory also offers a bridge to financial and institutional applications. Due diligence, credit committee research, regulatory analysis, corporate restructuring, and compliance investigations all involve overlapping evidence domains and contingent lines of inquiry. A fixed retrieval pipeline may be sufficient for routine questions. An adaptive research agent becomes potentially valuable when the next relevant question is not known until the previous document has been examined.

This suggests a broader thesis: the distinctive contribution of an autonomous knowledge architecture is not that it holds more documents, nor that it produces longer answers. It is that it can allocate attention among unresolved questions, gather evidence selectively, and revise its investigative plan. The design challenge is to grant enough discretion for meaningful adaptation while retaining the boundaries that make conclusions auditable and actions governable.

Design consideration: Institutional accountability

The importance of institutional accountability becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate institutional accountability into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, institutional accountability should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Research discretion

The importance of research discretion becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body

of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate research discretion into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, research discretion should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes. Interpretation should begin with the path, not the polished memorandum. For each consultation, reviewers can ask whether the system found the expected critical records, whether it pursued useful cross-domain bridges, whether an exception or missing predicate changed its plan, and whether it stopped for a defensible reason. A high recall score can coexist with poor citation fidelity if documents were found but attached to unsupported propositions. A low call count can indicate efficiency or premature termination. Reporting a small set of trace examples alongside aggregate metrics lets the reader distinguish these mechanisms. The comparison should also identify failure categories—missed seed, incorrect metadata filter, misleading edge, repeated search, unsupported synthesis, or budget exhaustion—because different causes require different remedies.

The experiment cannot establish that the agent understands Mexican law, that its fictional corpus represents the full complexity of legal practice, or that adaptive navigation will outperform simpler methods on every question. It can test narrower claims: whether the policy adapts its next action to new evidence, whether it can retrieve cross-domain material under a limit, and whether its resulting claims are better grounded than those of defined baselines. The system should be judged against these declared constructs rather than a vague notion of intelligence. If performance changes when one fact is varied, the trace should reveal whether the change follows the altered predicate. If performance does not change, that too is informative and may reveal that the agent ignores context. Students can use these observations to separate navigation competence from writing fluency and to recognize that an auditable process is valuable even when the model's recommendation remains uncertain.

Implementation implications

The next step is not immediate deployment but a staged validation sequence. First, implement the synthetic corpus and deterministic baselines so the data model, relation semantics, and scoring key can be inspected without a model. Second, add the autonomous policy behind a restricted executor and compare its traces against those baselines. Third, ask independent reviewers to examine claim support, treatment of exceptions, and missing-fact notices. Fourth, test the system under deliberately difficult conditions such as stale materials, ambiguous facts, disconnected graph paths, malformed model outputs, and exhausted budgets. Only after these stages should the team decide whether a supervised pilot is justified. Each gate should have written acceptance criteria and a named decision-maker.

A successful demonstration would establish a limited proposition: within a defined synthetic environment, an agent can choose among bounded evidence-seeking actions and produce a trace that a reviewer can reconstruct. It would not establish that the agent understands Mexican law, that its generated analysis is correct for real cases, or that autonomy improves professional outcomes. Conversely, a failed demonstration can still be informative if the trace shows whether the failure arose from corpus defects, relation modeling, tool execution, policy selection, output synthesis, or evaluation design. The architecture is valuable in part because it makes these failure locations distinguishable.

A later multi-agent laboratory should be conditional on these results. Multiple agents may investigate separate domains, challenge assumptions, or search for contrary authority, but coordination also creates risks: duplicated evidence, inconsistent source interpretation, emergent overconfidence, hidden responsibility, and higher review costs. Before adding agents, the single-agent experiment should show reliable provenance, bounded actions, calibrated uncertainty, and an effective human escalation path. The next research question should concern whether specialization

and disagreement improve evidence quality enough to justify their cost, not whether several models can produce more text.

The enduring contribution of the Second Brain idea is therefore institutional rather than merely computational. External memory can make evidence more available; typed relationships can make connections inspectable; bounded navigation can make research priorities adaptive. None of these features removes the need to state who is responsible for a conclusion and on what basis it may be defended. If the system makes sources, assumptions, uncertainty, and limits easier to examine, it can support professional work. If it hides those elements behind fluency or automation, its apparent intelligence becomes a liability. Further development should be judged against that standard rather than against novelty alone.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Gate development from synthetic prototype to supervised pilot using written criteria for provenance, claim support, uncertainty, and escalation.
- Preserve failure traces that identify whether a defect came from data, graph structure, tools, policy choice, synthesis, or scoring.
- Require independent review of exceptions, contrary evidence, missing-fact notices, and source-to-claim mappings before progression.
- Defer multi-agent coordination until single-agent limits, responsibility assignments, and reviewer controls have been demonstrated.
- State the narrow proposition each evaluation supports and prohibit claims of legal competence or real-world accuracy beyond that evidence.

11 Conclusion and next laboratory

SECTION OVERVIEW

This conclusion consolidates the claim that autonomous research is adaptive evidence selection under inspectable limits. It identifies what the synthetic experiment can establish, what it cannot establish about real legal practice, and how a later laboratory could test transfer and robustness.

The proposed Level 3 experiment completes a clear pedagogical arc: from independent notes, to interconnected evidence, to endogenous navigation over a multidimensional vault. Its synthetic Mexican legal setting is deliberately demanding. Constitutional foundations, specialized legislation, administrative processes, jurisprudential interpretation, and cross-border implications create many possible routes through the same information space. A competent research system must discriminate among them rather than simply traverse every available edge.

The laboratory is also designed to resist the temptation to equate complexity with success. More documents and more graph layers can make retrieval harder; an LLM can waste steps, follow irrelevant leads, or stop prematurely. The controlled comparison against semantic and deterministic baselines is therefore central. The research agent should earn its additional complexity by demonstrating improved evidence coverage, reliable citations, sensible stopping decisions, and reasonable cost.

The most important practical design choice is the separation between judgment and execution. The language model proposes the next research action; a tool gateway checks the request and returns bounded evidence; the research state is updated; and an evaluation layer later tests whether the resulting memorandum is defensible within the fictional corpus. This architecture can be explained to nonprogrammers without hiding the operational reality. Students can see exactly which source was visited, which relation was followed, which issue motivated the action, and why the agent eventually stopped.

Finally, the experiment prepares a natural Level 4. Once one agent can navigate a complex vault, several specialized agents might investigate different legal domains, challenge each other's assumptions, and coordinate under a supervisory protocol. That future step should not be taken prematurely. Level 3 must first establish that a single autonomous navigator can use judgment to improve the process of evidence acquisition. The proposition to be tested is concise: *an autonomous Second Brain does not merely retrieve what it knows; it decides what it needs to know next.*

Design consideration: Scope and limits

The importance of scope and limits becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate scope and limits into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, scope and limits should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

Design consideration: Empirical validation

The importance of empirical validation becomes apparent when the researcher confronts a question whose answer depends on several different documents. In this laboratory, a plausible sentence is never sufficient evidence of a legally defensible conclusion. The system must identify the relevant proposition, its documentary source, the conditions that make it applicable, and the remaining factual or interpretive uncertainty. This is especially important in a Mexican-law teaching environment: students must distinguish constitutional principles, legislation, subordinate regulatory instruments, administrative acts, and judicial interpretation without collapsing them into one undifferentiated body of text. The synthetic setting makes these distinctions observable while avoiding the false appearance that invented provisions are authentic Mexican law.

Operationally, the proposed Colab notebook should translate empirical validation into inspectable data and decisions. Every research step needs a documented purpose, an authorized tool call, a source identifier, and an observable outcome. A result that changes the research plan should record why it did so; a result that does not change the plan should still be available for inspection. Students should be able to reconstruct the path from initial consultation to final memorandum and ask whether a different path would have yielded stronger evidence. The agent may select a promising route, but it must not silently create a legal rule, conceal a missing source, or treat a speculative inference as a verified proposition.

From a methodological perspective, empirical validation should also appear in the evaluation rubric. We should reward relevant discoveries, calibrated uncertainty, accurate source use, and the identification of genuine gaps rather than sheer document volume. A shorter path may be better than a longer one when it reaches the decisive provisions; a longer path may be justified when an exception or conflicting authority requires investigation. These distinctions are precisely why endogenous navigation deserves to be tested against simpler retrieval baselines. The experiment should expose both its potential advantages and its characteristic failure modes.

The next laboratory should preserve the discipline developed here while broadening the evaluation target. A follow-on study could compare different corpus topologies, vary the proportion of superseded and exception-bearing records, or test whether a human researcher benefits from seeing the agent's trace before composing a memorandum. Each extension should begin with a stated hypothesis, a baseline, and an outcome that could count against the hypothesis. A larger corpus is not automatically a better one; it is useful only if its added complexity creates questions that the evaluation can resolve. Likewise, adding tools should be justified by a research need and matched by controls that keep their authority and side effects visible.

The durable lesson is therefore procedural. A second brain becomes a research partner only when it can help expose what is known, what is connected, what is missing, and why a next step is justified. The model's autonomy lies in selecting among permitted investigative moves, while humans remain responsible for defining the question, assessing the significance of sources, and deciding whether the evidence is adequate for any real-world use. The proposed Level 3 experiment turns that division of labor into a testable design: synthetic records create controlled conditions; typed graphs make relations inspectable; bounded actions permit adaptation; and evaluation rewards grounded coverage, honest uncertainty, and stopping discipline. Those features provide a foundation for further teaching experiments without converting a demonstration into a claim of legal authority or completed empirical success.

Implementation implications

The three-level curriculum offers a practical sequence for deciding whether a research system is ready for greater discretion. A Level 1 system should first show that retrieval returns identifiable passages rather than merely similar language. A Level 2 system should show that links are typed, reviewable, and useful for navigating related materials. Level 3 should be considered only after those foundations are stable, because an adaptive policy cannot compensate reliably for ambiguous source identity or defective relationships. Each level should preserve its own baseline and documented acceptance criteria. This staged approach prevents the novelty of autonomous navigation from obscuring whether the underlying information environment is fit for use.

For an organization considering a real implementation, the principal decision is not whether an agent can produce a plausible memorandum. It is whether the organization can specify the task, supply current and authorized materials, protect confidential information, assign a reviewer, and reconstruct the path from question to conclusion. If any of these conditions is absent, the appropriate output may be a limited research aid or a recommendation not to automate the workflow. An accountable institution should also define how it will respond when the agent identifies a possible issue that falls outside the original mandate. Escalation should be available without treating the agent's issue label as a legal determination.

The experiment should therefore be judged by the quality of its process as well as by its answer. A useful outcome

may be a clear list of unresolved facts, a warning that an authoritative source is missing, or a reasoned refusal to conclude. These are not failures when the evidence is genuinely insufficient. Conversely, a concise answer is not a success if it hides exceptions, cites irrelevant records, or omits a consequential uncertainty. Evaluation should reward the disciplined handling of incomplete evidence and should report the costs of verification. This standard aligns the paper's central proposition with professional practice: autonomy has value only when its exercise remains bounded, reviewable, and correctable.

Future development should proceed through evidence gates rather than a fixed promise of increasing automation. The next release can document the synthetic baseline and test whether independent reviewers reproduce the same claim-support judgments. A later pilot, if justified, should use a narrow purpose, a defined set of source types, and a human approval step, with a pre-established route to suspend the system. Broader deployment should require evidence that changes in source versions, fact patterns, and user behavior do not erase the benefits observed in controlled conditions. The decision to expand scope should be recorded with the evidence considered, dissenting views, and residual risks. This makes future work an accountable continuation of the laboratory rather than an assumption that technical progress automatically warrants greater delegation.

REAL-WORLD CONCLUSION CHECKLIST

- Advance from retrieval to graph use to autonomous navigation only after each stage meets documented provenance and review criteria.
- Define an organizational owner, narrow approved purpose, accountable reviewer, and suspension authority before any real-world pilot.
- Treat qualified uncertainty, missing-source alerts, and justified refusal as potentially successful outcomes when evidence is incomplete.
- Record expansion decisions with evaluation evidence, reviewer disagreement, known limitations, and the residual risks accepted.
- Reassess the system whenever sources, models, prompts, user groups, or intended uses change; preserve comparisons to prior versions.

12 Integrated implementation: from theory to the sixteen-cell laboratory

The original monograph established the theoretical argument and an initial ten-cell blueprint. The executable laboratory subsequently evolved into sixteen code cells. This chapter documents that progression using the uploaded notebook as the primary implementation source. The descriptions below concern the code's design and observable operations; they do not assert that the live model path or every export has succeeded in an independent Colab execution.

READING THE LABORATORY

The tutorial proceeds from corpus construction to multilayer relationships, retrieval tools, legal consultations, baselines, autonomous navigation, evidence synthesis, evaluation, dossier export and committee-style writing. Cells 7 and 8 are the conceptual hinge: the former initializes the evidence state; the latter chooses which connected source to examine next.

12.1 Cell 01 — Environment, reproducibility and model configuration

The first cell establishes the computational and documentary environment in which every later cell will operate. Its responsibilities include importing libraries, integrating Google Drive, selecting a root directory, creating output folders, configuring optional Claude access, and declaring an explicit model identifier. These choices establish the conditions under which a run can be reproduced; they do not themselves show that an experiment has succeeded. Centralizing paths is a research integrity decision rather than a convenience. The canonical corpus, generated Obsidian notes, agent traces, evaluation tables, and committee deliverables should have predictable locations, so that later cells neither overwrite unrelated files nor silently read stale outputs. The root should be chosen in one place and referenced consistently, with directory creation handled before any output is written. The configuration should also make clear which services are optional and what happens when credentials or network access are unavailable. A model name written in a notebook is only a requested setting. The identifier requested for this project is Claude Haiku 5.5, but the provider must accept that identifier and return a response before the notebook can report that the live model path works. Each run should record the configured identifier, actual provider identity when returned, execution time, and whether live access or fallback behavior was used. Credentials belong in a protected secret store or runtime prompt, never in saved cells, printed diagnostics, or exported dossiers. The deterministic fallback should be visibly distinguished from a model judgment in both traces and later analysis. It is useful for teaching the deterministic portions of the workflow, but its choices must not be attributed to the language model. A clear environment summary can report imports, paths, available folders, configuration status, and missing optional settings without exposing secrets. It should avoid implying that an API call has been tested merely because configuration succeeded. Likewise, a successful setup should not be described as proof that every later export will work. Students should inspect the summary before continuing and resolve path or credential warnings deliberately. By isolating setup, the notebook makes environmental assumptions easier to review and rerun. This cell therefore provides the controlled starting conditions, documents the execution mode, and prepares a clean boundary between local reproducible work and external model services. Before proceeding, the operator should confirm that inputs, directories, and execution mode match the published run record. Record package versions, runtime date, and storage permissions for later comparison.

12.2 Cell 02 — Synthetic legal corpus: documentary ontology

Cell 2 constructs the experimental evidence base: 350 entirely fictional documents distributed across ten domains. The allocation is 25 constitutional, 40 civil, 45 commercial, 35 criminal, 45 tax, 35 administrative, 35 securities, 25 customs, 30 international, and 35 jurisprudential records. Every record receives a stable identifier, a domain, a document type, two topical labels, a synthetic status, and a short text. The identifier is the durable key used by retrieval, graph edges, traces, citations, and exports; it must not change when a note is reformatted. Status flags distinguish illustrative current, exception sensitive, and superseded fictional material. These labels model the general research problem of applicability, including temporal and procedural qualifications, without representing real Mexican legislative history or offering statements about operative law. The labels should be defined consistently so students can tell whether a source is merely illustrative, qualified by an exception, or intentionally obsolete within the exercise. The corpus should be generated deterministically or saved after construction, then written once as a canonical JSON file. Later graph, indexing, navigation, and evaluation procedures should use that same

evidence base rather than recreating documents independently. Otherwise, a baseline and an agent could be judged against different source populations, making comparisons invalid. A schema check should confirm that all records have required fields, identifiers are unique, domain values belong to the declared set, and texts are nonempty. The instructor should inspect representative records from every domain and status category before students interpret metrics. The allocation is intentionally uneven enough to resemble a multidisciplinary collection, but document counts alone do not imply importance or legal authority. A domain with more records may appear more often in retrieval simply because it is more densely represented. That imbalance should be visible in summaries and considered when discussing recall. Controlled corpus construction lets the instructor distinguish evidence availability from navigation failure. If a method misses a target document that exists in the corpus, the issue concerns retrieval or path selection; if the corpus lacks a necessary proposition, no agent can recover it. Students should therefore treat this cell as the boundary of what the experiment can know. Its output is a transparent synthetic dataset suitable for controlled comparison, not a substitute for statutes, cases, official publications, or advice. This review also gives students a fixed reference for discussing omissions, duplication, and representativeness before evaluating any agent. Inspect examples across domains and status before sharing the dataset.

12.3 Cell 03 — Five-layer graph and Obsidian-ready legal vault

Cell 3 changes the knowledge base from a document collection into a connected research environment. It represents records as nodes in a NetworkX multigraph, allowing the same pair of documents to carry multiple distinct relationship types without collapsing them into one generic link. Five layers are encoded: hierarchy, substantive, interpretation, procedure, and cross border. In the teaching corpus, these labels help students ask different questions about why two records might be examined together. A hierarchy cue can direct attention to an apparent relation between levels of instrument; a substantive cue connects shared topics; an interpretation cue points toward an explanatory or jurisprudential record; a procedure cue highlights a process relationship; and a cross border cue makes an international dimension visible. Links are generated through reproducible intra domain rules, constitutional and jurisprudential connections, thematic bridges, and selected domain pair bridges. The precise rule and its inputs should be inspectable so the graph can be regenerated and its structure audited. Importantly, every edge is a synthetic research cue. It does not establish legally valid authority, normative precedence, binding force, or actual applicability to any real dispute. A visually prominent connection is not a legal conclusion, and the graph must not turn similarity into authority by implication. Once the graph exists, each record is exported as a Markdown note containing YAML metadata and typed Obsidian wikilinks. The metadata should preserve the stable identifier, domain, document type, status, and topics used by retrieval and evaluation. Typed links make relationships readable in the vault while retaining their conceptual layer. Exporting after graph construction ensures that notes include the complete relationship structure rather than a partial set of links created before all edges exist. A graph manifest should list nodes, edge endpoints, relationship layers, and generation details in a form that can be checked independently of the visual vault. Students can compare a note's outgoing links with the manifest and investigate missing, duplicated, or unexpected edges. Stable identifiers also allow the notebook and Obsidian representation to be compared without relying on titles that may change. The resulting vault serves both as a human readable research interface and as an inspectable export of the computational graph. This cell therefore demonstrates how externalized knowledge can encode relationships while preserving an explicit warning that synthetic connectivity is not legal authority. Record generation settings alongside the manifest so another reader can recreate the same nodes and edges. Review exported links.

12.4 Cell 04 — Semantic indexing and graph-aware research tools

Cell 4 constructs the retrieval interface used by every subsequent research method. A TF IDF vectorizer indexes each document's short text and assigned topics, using unigrams and bigrams, English stop word removal, and a vocabulary ceiling of 6,000 features. The precise feature settings matter because they define which expressions can influence similarity rankings. English stop word removal is a design choice that should be discussed when the corpus contains Spanish legal vocabulary; it may have little or unintended effect on particular terms, so the resulting rankings should not be treated as language independent. Cosine similarity compares a consultation representation with document vectors and returns a ranked list. This is a candidate generation mechanism, not a legal relevance determination. A high score indicates lexical or topical resemblance under this representation; it does not establish that a document is current, authoritative, applicable, or sufficient. The cell also defines direct document inspection, which retrieves the text and metadata for a selected stable identifier, and a graph frontier function, which lists unvisited neighbors of the evidence already examined. Each frontier item should expose its identifier, domain, semantic score, and graph relationship layers so that a researcher can understand why the candidate is available.

The retrieval and frontier procedures have different roles: semantic search proposes initial candidates, while graph navigation reveals additional connected records. Keeping them separate permits later methods to use the same search foundation but follow different research policies. It also makes failures easier to diagnose. A relevant document omitted from the initial ranking may reflect limitations in text or vectorization; a relevant neighbor that exists but is never selected may reflect navigation choices. The frontier should exclude visited identifiers and should not silently include nodes absent from the corpus. Empty queries, empty text fields, and ties should receive predictable handling. Search results should preserve their scores and stable ordering where possible, because reproducibility depends on understanding how candidate lists were produced. Students should inspect examples of high and low similarity results and compare those scores with the actual synthetic text. This cell supplies reusable tools rather than an autonomous researcher. Its output supports the baselines and gives the later agent a constrained menu of evidence to consider. The central methodological principle is that similarity scores can suggest where to look, but they cannot determine legal authority or decide which evidence the agent must pursue. Scores remain candidate signals, never legal findings.

12.5 Cell 05 — Three multidisciplinary legal consultations

Cell 5 supplies three consultations that turn the synthetic vault into a meaningful experiment. Case A concerns foreign investor financing, allegedly misleading disclosure, and contractual breach. Its intended research dimensions include commercial, securities, tax, administrative, and constitutional material. Case B concerns import classification, a customs declaration, and a tax assessment, with customs, tax, administrative, criminal, and constitutional dimensions. Case C concerns a foreign investor acquisition involving a share transfer, financing, and regulatory filings, spanning commercial, international, securities, tax, and civil topics. The cases are designed to require movement across domains rather than simple lookup of one keyword. Each consultation should state enough facts to motivate research while leaving relevant uncertainty unresolved, so the task is investigation planning rather than guessing a predetermined legal outcome. The facts and legal descriptions are synthetic and should be labeled as such wherever the cases are displayed or exported. For evaluation, the notebook selects one high scoring synthetic document per target domain as an instructor key. Those keys are withheld from the agent so that they can serve as a separate measurement reference instead of guiding the navigation process. Their function is limited: they indicate records the instructor expects a method to retrieve for the designed exercise. They are pedagogical labels, not an exhaustive gold standard, objectively authoritative legal answers, or proof that other sources are irrelevant. A system that retrieves the key may still misunderstand it, and a system that misses it may find another relevant record. The instructor should document why each key was selected and review whether the corpus actually supports the issue it represents. Target domains likewise express expected breadth for the scenario, not an instruction that every case must include every possible legal perspective. Students can compare the initial semantic results with the target domains and consider whether wording in a consultation creates bias. Cases should remain fixed across compared methods so differences in outcomes arise from research strategy rather than changed facts. The cell therefore establishes a controlled evaluation setting with visible assumptions, hidden measurement labels, and multiple plausible paths. Its design invites students to question what counts as coverage and how much confidence a small instructor key can support. It must never be presented as a simulation of real client advice or a declaration about actual Mexican law. The cases can be reviewed for balance before any method is run. This flags wording that favors one strategy initially.

12.6 Cell 06 — Non-agent baselines: retrieval and fixed traversal

Cell 6 provides two counterfactual methods against which autonomous navigation can be compared. The semantic baseline simply selects the highest ranked documents for a consultation, offering a reference for what direct similarity retrieval accomplishes without graph traversal. The fixed graph baseline begins from three semantic seeds and follows graph neighbors in a deterministic breadth first pattern, subject to defined depth and document budgets. It demonstrates how connections can broaden a search even when the traversal policy is not adaptive. Both methods use the same corpus, consultation text, retrieval index, and target cases as the autonomous system. Holding these inputs constant is necessary for interpreting differences as consequences of method rather than data variation. Their purpose is to make the possible contribution of agent judgment testable. Without baselines, a visually interesting path might be praised simply because it contains multiple domains, even if a simpler procedure would have produced equivalent evidence. The notebook should preserve each method's ordered path, visited identifier set, and trace, allowing evaluators to compare not only final document counts but also the route taken. Since the baselines make no language model decisions, they also show what can be achieved through deterministic rules alone. Equal document budgets support a fairer comparison of evidence acquisition, but do not make the methods equivalent in computational cost. A similarity ranking may be inexpensive, breadth first traversal may require graph operations, and a live agent may

consume API calls, latency, or tokens. Those costs should be reported separately rather than hidden behind a shared document limit. The fixed traversal's deterministic order should be documented, including how ties and multiple relationship types are resolved. If it follows arbitrary implementation order, repeated runs may not provide the intended stable comparison. The baselines are not straw figures: they should be competently implemented and given access to the same permissible evidence. Results may show that a simple method performs as well as or better than the agent, which is a meaningful finding rather than a failure of the exercise. This cell gives the study its comparative structure and helps students separate novelty from demonstrated value. It supports disciplined evaluation of whether adaptive selection improves coverage, relevance, or efficiency, while recognizing that no synthetic benchmark can establish universal superiority. This design also permits students to compare paths without conflating deterministic traversal with adaptive evidence selection. Report any ordering rule that could change the baseline path for fairness.

12.7 Cell 07 — Initial evidence: Cell 7 as the research seed

Cell 7 initializes the research state from the three highest ranked semantic documents returned for the consultation. These seed records establish the agent's initial field of view; they do not prescribe the complete path or imply that the first results are the most authoritative evidence. The state should retain the original question, visited identifiers, ordered path, decision trace, unresolved issues, encountered errors, and model call counter. Keeping these elements together gives the next cell a defined account of what has already been inspected and what remains uncertain. It also allows the final report to reconstruct the progression from consultation to evidence set. The seed count is a deliberate parameter. A small initial set can conserve budget and force the agent to use connections, but it can also overemphasize a narrow interpretation of the question. A larger set may improve early coverage while reducing the importance of adaptive exploration. The notebook should therefore make the seed rule visible rather than burying it in helper code. Since the seeds come from semantic ranking, their selection may reflect vocabulary overlap, assigned topics, document density, or phrasing choices. A consultation that names a domain directly may produce a different starting state from one that describes the same issue indirectly. Students should inspect seed metadata and compare it with the scenario's target domains before drawing conclusions about later behavior. This cell represents the first half of the conceptual transition from retrieval to research. A conventional search engine can suggest where to begin; it cannot by itself determine what evidence ought to be acquired next after new information appears. The stored state gives the agent room to make that next choice while preserving the evidence and questions that inform it. It should also distinguish the initial seed list from later visits, so evaluators can identify whether domain breadth came from initial retrieval or subsequent navigation. If a seed is missing or malformed, the state should record that condition rather than silently substituting a different source. The output is an initialized research agenda, not a conclusion, legal memorandum, or declaration of sufficiency. By making the starting conditions explicit, Cell 7 allows the following policy to be assessed on its actual incremental contribution. Students can ask whether the agent corrects seed bias, compounds it, or simply follows the strongest available connection. That question anchors the laboratory's study of endogenous navigation. Attribute later coverage to seeds or navigation separately.

12.8 Cell 08 — Autonomous research: Cell 8 as a bounded judgment engine

Cell 8 is the core of Level 3 because it turns accumulated evidence into a constrained choice about what to inspect next. At each iteration, the system constructs a candidate frontier from graph neighbors of the records already visited, excluding identifiers that have been inspected. It then supplies the consultation, selected inspected document metadata, unresolved issues, available candidates, and remaining budget to the configured language model. This context is intended to support a research action, not to invite an unconstrained legal opinion. The model may return only a structured JSON decision: visit one offered identifier or stop, accompanied by a short explanation of the stated research purpose. The execution layer parses the response, validates the action type, confirms that the proposed identifier is in the offered frontier, and checks the applicable limits before accepting it. Malformed output, an unknown identifier, an API error, or unavailable model access must not silently become an accepted model decision. Instead, the code invokes a deterministic heuristic where appropriate and records that fallback explicitly. Minimum evidence requirements, a maximum document budget, and a cap on model calls constrain the loop; the stop action should also be assessed against the sufficiency audit in Cell 12. Every accepted visit updates the ordered path, visited set, trace, and remaining budget. Each trace entry should preserve the iteration, selected action, target, reason, execution mode, tool result, and any resulting change in open issues. This makes the agent's observable behavior available for review without treating its stated rationale as proof of sound reasoning. A robust implementation should handle empty frontiers, repeated identifiers, provider timeouts, invalid JSON, and exhausted budgets with explicit termination states. Those conditions are part of research operations and should be measured rather than hidden. The deterministic fallback is operationally useful for demonstrating the loop and maintaining a

runnable notebook when model service is absent. However, fallback choices are generated by a different policy and must never be reported as live Claude judgments or combined with live results without labeling. Students should compare traces across modes and note that a plausible reason may be generated after a choice without faithfully representing an internal deliberation. Cell 8 therefore demonstrates bounded autonomy as a system property: the action space is narrow, constraints are executable, decisions are logged, and a human reviewer can inspect how evidence was acquired. It does not establish professional legal reliability, source authority, or autonomous legal judgment.

12.9 Cell 09 — Structured evidence memorandum

Cell 9 converts the completed research path into a first written artifact: a structured evidence memorandum. It records the consultation, research mode, documents inspected, domains encountered, selected fictional source identifiers, applicability flags, and facts requiring human verification. The purpose is to make the evidence state readable before asking a system to produce a developed narrative. A reviewer should be able to connect each listed source to a stable corpus record and to determine whether it was actually inspected during this run. The implementation checks that bracketed synthetic citations refer only to identifiers in the visited set. It can catch a reference to an unvisited or nonexistent record when identifiers are well formed; it cannot establish that a cited passage supports the nearby proposition, that the record is correctly interpreted, or that the proposition is legally accurate. Nor does a visited source become authoritative merely because it appears in the memorandum. The report should distinguish the consultation's supplied facts from assumptions, generated issue labels, and open questions. It should also identify whether the research used live model decisions or a deterministic fallback, since those execution modes represent different processes. Domain summaries can reveal whether exploration remained within a single field or crossed the intended boundaries, while applicability flags help students notice status distinctions that require further review. The memorandum is intentionally closer to a research inventory than a committee ready legal opinion. Its headings and concise fields support comparison across methods without suggesting that a polished format resolves uncertainty. It provides the controlled evidence package from which Cell 16 may later prepare a more developed synthetic memorandum. That downstream drafting cell should receive only the evidence actually inspected, not hidden evaluation keys or unvisited records. Instructors can use discrepancies between this inventory and the trace to identify implementation errors, and can ask students whether each recorded issue has a corresponding source or remains a factual question. If an evidence field is empty, the output should say so rather than inventing content. When a citation audit rejects an entry, the rejection should be visible for correction. A saved copy should retain the case identifier, run identifier, date, method, and source list so it remains interpretable outside the notebook session. The artifact supports accountability and human review, not independent legal advice. This cell therefore marks the transition from navigation data to a structured account of what the process has and has not found.

12.10 Cell 10 — Comparative research evaluation

Cell 10 evaluates the research methods using a common set of criteria, including documents visited, target domain coverage, instructor key recovery, topical breadth, and model call consumption. Metrics should be calculated from saved run data, not copied from a schematic image or estimated from memory. The research question is not simply whether the autonomous agent inspects more notes, because methods may share the same document budget and a larger set is not automatically more useful. Instead, the comparison asks whether adaptive navigation uses its available budget to assemble a more relevant, diverse, and reviewable evidentiary record. Each metric captures only one aspect of performance. Visited count measures volume; domain coverage measures breadth relative to the case design; key recovery measures whether selected expected records were found; topical breadth reflects the variety of assigned subjects; and model calls indicate one component of resource use. None alone measures legal correctness, citation support, reasoning quality, or whether the investigation should have stopped. The notebook's scoring is experimental. Instructor selected key documents provide a transparent teaching proxy for comparison, not a judicial determination of which sources would control a real dispute. Their limited size means that a method may find useful nonkey documents without receiving credit, while recovering a key does not show that the system understood it. Evaluation should report raw counts and denominators alongside normalized scores, so a reader can see how coverage was calculated. Any weighting scheme should be stated before interpreting results, since different weights can change which method appears preferable. Where repeated runs are available, report variation rather than selecting a favorable run. Live model and fallback outcomes must be separated or explicitly stratified, as they do not represent the same decision process. Computational cost should also be treated carefully: equal document budgets do not equalize API latency, token consumption, or local processing effort. A fair comparison preserves case wording, corpus version, starting seeds, and evaluation key across methods, while recording any failures or

incomplete executions. Results should be reproduced from the user's actual run rather than inferred from illustrative numbers in an infographic. Students should ask what each metric omits and whether the combined measures answer the research question. Cell 10 makes performance claims inspectable and falsifiable, but it cannot turn synthetic scores into evidence about real legal practice. Its main contribution is a disciplined comparison that can reveal gains, tradeoffs, or no measurable advantage for adaptive navigation.

CELLS 10–14: FIVE COMPLEMENTARY TESTS

Cell 10 compares methods; Cell 11 inspects provenance and domain coverage; Cell 12 audits evidence sufficiency; Cell 13 tests research-budget sensitivity; and Cell 14 examines robustness and traces. The accompanying five-panel infographic should be read as an overview of these distinct diagnostic stages.

12.11 Cell 11 — Provenance, legal domains and authority cautions

Cell 11 examines the source inventory as a research governance problem rather than treating retrieval as the end of inquiry. It assembles metadata concerning domains, source types, status labels, identifiers, and evidence coverage, making visible the difference between finding a thematically relevant note and determining whether that note is appropriate to rely upon. In a genuine legal research environment, source evaluation would require checking the issuing body, official publication, amendment history, effective dates, jurisdiction, procedural posture, and any applicable rules concerning legal force. Constitutions, statutes, regulations, administrative instruments, judicial decisions, and contracts serve different functions; similarity scores or graph edges cannot reliably rank them. In real Mexican practice, each category must be verified through current authoritative sources and interpreted in context by qualified professionals. The synthetic corpus does not perform those real world verifications. It represents distinctions through metadata and warnings only, and its labels are teaching constructs rather than validated authority classifications. The cell should present counts by source type and domain while preserving the status of each record. It can show illustrative or superseded items, or reveal an omitted target domain. Such observations help the class discuss provenance and applicability as explicit research tasks. They do not prove that a source controls, that a particular hierarchy has been accurately modeled, or that a real rule applies to a hypothetical fact. Students should treat metadata as a prompt to verify, not as a substitute for verification. The audit can also reveal structural imbalance in the corpus, such as domains with many more records or source types with few examples. That imbalance can affect retrieval and coverage metrics, so the results should be interpreted against corpus composition. Where provenance fields are missing or inconsistent, the notebook should surface the deficiency rather than silently infer a value. A useful output links each summary figure to the underlying record identifiers, allowing a reviewer to inspect how the count was produced. The cell teaches a repeatable discipline: identify what a record claims to be, establish where it came from, determine what status information is available, and mark what remains unverified. It distinguishes a searchable fictional record from evidence suitable for reliance. This is particularly important in a legal research laboratory, where authentic looking titles or institutional labels might otherwise be mistaken for actual Mexican law. Cell 11 models caution and auditability; it does not certify legal authority or reproduce Mexico's operative legal hierarchy.

12.12 Cell 12 — Sufficiency and stopping audit

Cell 12 independently audits whether the research process stopped after reaching a declared minimum evidence threshold and a specified level of domain breadth. This check is important because the agent's own statement that it has enough information is not a reproducible measure of sufficiency. The audit applies explicit conditions outside the language model's narrative, such as a minimum number of inspected records, required target domains, unresolved issue count, or completion of a specified evidence checklist. These criteria must be visible and appropriate to the synthetic case. They are instructional operational definitions, not universal rules for legal research or guarantees that an answer is correct. Conversely, an agent can exhaust its document budget without resolving important factual or interpretive questions. The audit should distinguish these outcomes rather than treating every termination as equivalent. It should record the stop action, the model's stated reason if available, whether the choice came from live model or fallback behavior, the visited identifier set, achieved domain coverage, remaining budget, and open questions. It can then compare the stated rationale with the external criteria: for example, whether the system claimed broad coverage while omitting a required domain, or whether it stopped with unresolved exceptions. A failed sufficiency check should not automatically cause the system to continue if no budget remains; instead, the output should report that the search ended without meeting the threshold. A successful check likewise means only that the configured checklist was satisfied. It does not establish that the evidence is accurate, current, complete, or legally controlling. Instructors should explain why a threshold was chosen and examine how changing it affects

conclusions about efficiency. A strict breadth rule may penalize a focused investigation, while a permissive rule may let an incomplete path appear successful. Sensitivity to reasonable alternative thresholds is therefore informative. This cell also helps distinguish efficient stopping from premature closure, two conditions that can otherwise look identical in a final memorandum. Reviewers should inspect the actual records and open issues, not rely solely on a green status label or aggregate score. The output should preserve both passes and failures in the dossier for later comparison. Cell 12 thus places a deterministic audit beside adaptive judgment: the agent may propose when to stop, while a transparent external test shows whether that choice met the laboratory's announced criteria. The audit supports discussion of sufficiency; it is not a substitute for qualified review or professional responsibility.

12.13 Cell 13 — Research-budget sensitivity

Cell 13 varies the maximum number of inspected documents to examine how research outcomes change as the evidence budget expands. This sensitivity analysis matters because autonomous systems operate under constraints such as time, token allowance, API cost, client urgency, and reviewer capacity. A small budget may reward strong seed selection and efficient navigation, but it can also prevent discovery of an exception or a necessary cross domain source. A larger budget gives the agent more opportunities to broaden coverage, while also creating more chances to collect redundant or marginal material. The relationship between budget and usefulness may therefore be nonlinear. A useful analysis plots target domain coverage and instructor key recovery against the document limit, while reporting model calls and execution time separately. It can additionally display the growth in unique topics, open questions, and total retrieved records to show whether added work changes the evidence picture or merely increases volume. Comparisons should keep the consultation, corpus, seed method, and scoring rules stable. If model behavior is stochastic or provider settings vary, repeated runs at each budget provide a more informative picture than one trace. The reported budget must also be defined precisely: it may count documents inspected, model decisions, or both, and these are not interchangeable resources. A method that uses fewer model calls but inspects the same number of records has a different cost profile from one that reduces both. The analysis should account for the initial seeds when calculating total inspected documents, avoiding a misleading distinction between seed acquisition and later navigation. Where a budget increase produces no additional coverage, the result may indicate that the graph frontier is exhausted, the policy stops early, or new records are redundant. Where coverage rises but key recovery does not, additional breadth may not correspond to the chosen evaluation target. Such patterns are useful prompts for examining graph design and key selection. No universal budget fits every question, corpus, risk, or available review capacity. Navigation and budgets require joint design. Students should consider the marginal value of each additional inspection and whether stopping criteria adapt to new evidence. Budget sensitivity can reveal where an agent is brittle, efficient, or wasteful, but the synthetic experiment cannot estimate real professional costs or guarantee that a given limit is safe. The cell makes resource tradeoffs visible so that future system designs can specify them deliberately instead of treating a default as self evident.

12.14 Cell 14 — Robustness, perturbations and research trace

Cell 14 studies how navigation responds to variations in consultation wording and makes the resulting research trace available for inspection. Perturbations may include paraphrases, reordered facts, changed emphasis, omitted keywords, or distractor details that should not materially redirect the investigation. The point is not to expect identical paths for every wording. Different phrasing can reasonably surface different initial seeds, and a changed fact may alter which evidence is relevant. Instead, the exercise asks whether the system preserves attention to the central issues, reaches essential domains when warranted, and signals uncertainty when a perturbation makes the question ambiguous. Each trace should record the step, visited node, source of the decision, selected action, short reported explanation, execution mode, and any issue or budget change. A reviewer can compare those records across variants and identify where wording changes triggered a new path. The comparison may expose lexical dependence in semantic retrieval, sensitivity to distractors, inconsistent stopping, or reliance on one graph bridge. It can also show appropriate adaptation when the facts genuinely change. Separate harmless wording changes from factual changes. A trace makes the software's observable sequence inspectable, but it does not reveal every internal model process or prove that the reported reason caused the selection. A concise rationale may be post hoc, incomplete, or persuasive without being correct. Traces are evidence of what the program selected and what explanation it returned, not proof of valid legal reasoning. Instructors should avoid treating a stable path as inherently reliable: the system may consistently follow a misleading synthetic edge. Similarly, path differences are not automatically failures if the evidence remains adequate and the variation is explainable. Robustness assessment benefits from reviewing both quantitative outcomes and selected examples of the actual documents visited. Compare coverage, key recovery, stopping, and trace length across prompts. Any model or configuration change should be recorded because

it can alter behavior independently of the test wording. Robustness testing is thus a diagnostic instrument for instructors and reviewers, not a certification of professional legal reliability. It helps identify where further controls or human review may be needed. Results should be reported with limitations, especially because three synthetic cases cannot represent the diversity of real legal consultations. Cell 14 teaches that repeatability and adaptability need examination together: a system should be responsive to meaningful facts while resisting irrelevant phrasing changes, and its trace should allow a human to challenge the route it took.

12.15 Cell 15 — Research dossier and export

Cell 15 packages the experiment's outputs into a reproducible research dossier organized around each consultation. The dossier includes the case definition, visited records, navigation trace, evidence memorandum, diagnostics, and visualizations. A useful dossier preserves not only the final outputs but also the path and settings needed to understand how they were assembled. This includes stable source identifiers, corpus version or manifest, run identifier, method, model execution mode, budget settings, open questions, and relevant evaluation results. Consistent names and folders help reviewers compare cases and find missing artifacts. The export process should avoid silently replacing earlier runs; versioned or uniquely named output folders preserve the ability to reconstruct differences. Where a file is not produced because a dependency is unavailable or a cell fails, the dossier should record that absence rather than implying completion. Export details vary by notebook version and dependencies. A schematic infographic communicates intended design, not proof that every pictured artifact was generated in a particular run. Before relying on an export, inspect the run directory and verify that each expected file exists, opens correctly, and contains the corresponding case's material. Check that identifiers in the memorandum match the source records and that the trace is consistent with the visited set. Verify plot labels and denominators. The dossier should also distinguish planned artifacts from observed outputs and live model results from fallback demonstrations. This is especially important when a figure or template resembles a completed deliverable but was not created by the actual execution path. Exported documents should preserve synthetic status notices so they cannot be mistaken for operative law or real client advice after being separated from the notebook. If PDF creation or Obsidian export depends on optional packages, the notebook should make that dependency explicit and provide a clear success or failure message. The package supports review and reproducibility while preserving its source corpus and code. It is a review package, not a signed, approved, or legally sufficient opinion. Students can use it to practice reconstructing an investigation, locating unresolved assumptions, and identifying where additional evidence would be required. Since Cell 16 was added after the original dossier architecture, reviewers should verify execution order and confirm that the writing output is included only when the relevant cell completed successfully. Cell 15 therefore closes the computational workflow by making its artifacts portable while preserving provenance, caveats, and the distinction between an intended design and an observed run.

DOSSIER DESIGN VERSUS OBSERVED OUTPUT

The figure describes the intended committee package. Before relying on any particular export, inspect the run directory and verify that the corresponding file exists and contains the expected material. In particular, the full writing cell was added after the original dossier architecture and should be checked for execution-order dependencies.

12.16 Cell 16 — Committee-style memorandum drafting and validation

Cell 16 extends the original fifteen cell sequence from evidence organization to developed writing. Its intended output is a separate, approximately 400 to 600 word committee style synthetic memorandum for each case. The document should organize the question presented, relevant supplied facts and assumptions, fictional legal framework, integrated cross domain analysis, risk assessment, unresolved questions, and proposed committee action. This structure asks the drafting system to synthesize the inspected evidence rather than merely list retrieved records. The drafting engine should receive only sources actually inspected by the research agent, together with the consultation and appropriate output instructions. It must not receive instructor hidden evaluation keys, which would leak the benchmark into the answer and invalidate the comparison. The input set should be recorded so reviewers can confirm that every cited identifier was available to the drafting stage. An identifier audit can reject nonexistent or unvisited citations, but cannot verify support or interpretation. The cell should distinguish live model drafting from a deterministic fallback summary, label the execution mode in the saved artifact, and report failures instead of presenting an empty or partial result as a finished memorandum. On success, this cell may export Markdown, PDF, and Obsidian notes. Formats retain synthetic status, case, run, sources, and open questions. Verify each export after execution. The proposed action should be framed as a recommendation for committee discussion inside the fictional scenario, not

as a real world legal instruction. The drafting prompt should discourage invented facts, unsupported authorities, categorical conclusions beyond the evidence, and concealment of uncertainty. Validation can check sections, length, citations, and evidence links; human reviewers must assess materiality, sound reasoning, tone, and factual fidelity in context. A polished layout can make weak analysis appear more credible, so presentation quality should never substitute for verification. The memorandum is educational and depends on a wholly synthetic corpus, constructed facts, and simplified evaluation rules. It is not legal advice and must not be applied as a statement of operative Mexican law. This final cell demonstrates a downstream use of an auditable research path while preserving the distinction between evidence collection and professional legal judgment. Reviewers should compare the final prose with the evidence memorandum and trace to identify omissions or unsupported transitions. If a source changes the risk assessment, the reasoning should explain that connection clearly. Cell 16 completes the laboratory's writing workflow, not the legal work that a real committee would require.

13 From the synthetic laboratory to Mexican legal practice

The teaching architecture offers a compelling institutional pattern, but an operational law-firm system would need a different corpus, a different legal authority model, and stronger safeguards. A real implementation should ingest authoritative texts with exact publication dates, amendment histories and pinpoint references. The system should distinguish the Constitution, statutes, regulations, administrative criteria, judicial decisions and contractual documents without treating their relative authority as a mere similarity score. It must record whether a proposition applies to the client's facts, the relevant period, the competent authority and the procedural posture. Privileged and confidential material requires access controls, data minimization and approved processing arrangements.

The research agent's decision power should be restricted to investigation planning and source inspection unless a firm has separately authorized additional actions. Human counsel should approve the issue framing, the evidentiary record, the legal interpretation and any committee recommendation. A committee-ready memorandum should identify the responsible lawyer, the status of source verification, material assumptions, dissenting interpretations, outstanding questions and the date through which the law was checked. The synthetic memorandum generated in Cell 16 intentionally does not satisfy those requirements.

13.1 A staged deployment protocol

First, validate retrieval and document provenance against a small set of official, dated sources. Second, test graph relations for false or misleading edges and add a mechanism for human correction. Third, evaluate the autonomous agent on expert-reviewed questions, including adversarial and out-of-domain prompts. Fourth, require legal review of citations and legal conclusions, not merely format and identifier checks. Fifth, monitor drift as legislation, jurisprudence, provider models and internal practice change. These stages turn an instructive experiment into a disciplined research-development agenda without overstating what the current notebook establishes.

14 Synthesis: the intellectual contribution of Level 3

The essential progression is from a passive repository to an active, bounded research process. A vault supplies documentary memory; the multilayer graph represents relationships; the semantic index proposes starting points; the agent selects a path through the evidence; deterministic controls enforce the permissible action space; and the memorandum and dossier make the resulting work inspectable. The architecture therefore places human and machine judgment in different roles. The model exercises constrained discretion over the next research step, while the institution remains responsible for the validity and use of the legal conclusions.

For teaching purposes, the contrast between Cells 7 and 8 should remain unmistakable. Cell 7 asks which documents initially resemble the consultation. Cell 8 asks which evidence ought to be investigated next, given what has already been learned and what remains unresolved. Cells 9–16 then test whether that research process can be explained, evaluated, packaged and translated into a disciplined written product. That is the defining contribution of this laboratory to the Second Brain sequence.

FINAL PEDAGOGICAL PROPOSITION

A Second Brain becomes more autonomous when it can choose its next evidence-acquisition action, justify that choice, operate within verifiable limits and leave a complete research record. The system does not thereby become an independent source of legal authority.

References and authoritative research framework

SECTION OVERVIEW

The materials below identify official Mexican legal research repositories and selected scholarship on legal analytics, retrieval, reasoning agents, hallucination, and AI risk. They support real-world verification and methodological context; none supplies rules for the fictional corpus.

1. Constitución Política de los Estados Unidos Mexicanos. Texto vigente, Cámara de Diputados, portal de Leyes Federales Vigentes. Consultative authority for constitutional structure; always verify the operative version.
2. Suprema Corte de Justicia de la Nación (SCJN). *Semanario Judicial de la Federación*. Official publication and search system for jurisprudence and judicial theses.
3. Suprema Corte de Justicia de la Nación. *Buscador Jurídico*. Official judicial research portal.
4. Cámara de Diputados, H. Congreso de la Unión. *Leyes Federales Vigentes*. Official consolidated legislative repository.
5. Diario Oficial de la Federación (DOF). Official gazette of the Mexican federal government; authoritative publication record.
6. Ley de Amparo, Reglamentaria de los Artículos 103 y 107 de la Constitución Política de los Estados Unidos Mexicanos. Verify current text in the official legislative repository.
7. Código Civil Federal. Verify current text in the official legislative repository.
8. Código de Comercio. Verify current text in the official legislative repository.
9. Código Fiscal de la Federación. Verify current text in the official legislative repository.
10. Ley del Mercado de Valores. Verify current text in the official legislative repository.
11. Ley Aduanera. Verify current text in the official legislative repository.
12. Ley Federal de Procedimiento Administrativo. Verify current text in the official legislative repository.
13. Ley General de Sociedades Mercantiles. Verify current text in the official legislative repository.
14. Ley Federal de Competencia Económica. Verify current text in the official legislative repository.
15. Susskind, Richard. *Tomorrow's Lawyers: An Introduction to Your Future*. 2nd ed., Oxford University Press, 2017.
16. Ashley, Kevin D. *Artificial Intelligence and Legal Analytics: New Tools for Law Practice in the Digital Age*. Cambridge University Press, 2017.
17. Lewis, Patrick et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." *Advances in Neural Information Processing Systems* 33, 2020.
18. Yao, Shunyu et al. "ReAct: Synergizing Reasoning and Acting in Language Models." *International Conference on Learning Representations*, 2023.
19. Ji, Ziwei et al. "Survey of Hallucination in Natural Language Generation." *ACM Computing Surveys* 55(12), 2023.
20. NIST. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1, 2023.

Source integrity note. These are real statutes, official research resources, and published scholarly works, not sources for the fictional rules of the laboratory. Their inclusion does not imply verification of the latest amendments or judicial criteria. All actual legal research must check current versions and the authoritative status of each source.

The official materials serve different purposes and should be consulted in a deliberate sequence. The Diario Oficial de la Federación provides the publication record for federal instruments and amendments; the consolidated legislative

repository offers a practical point of access to statutes, but a researcher should still verify the relevant text, reform history, and effective date against authoritative publication records. The Constitution supplies the framework for constitutional questions, while individual codes and statutes address narrower fields. A citation to a statute's title alone is not enough to establish that a provision governed a transaction on a particular date. Actual research should record the article or provision, the version consulted, the date of access, and the question for which it was retrieved.

Judicial materials require their own verification process. A search result, thesis heading, or excerpt should not be treated as a holding without examining the document, procedural context, issuing court, date, and status relevant to the question. The reference to the Semanario Judicial de la Federación and the SCJN's research portal is intended to direct future users to official research tools, not to imply that any particular fictional issue has a real controlling decision. When a question turns on interpretation, the researcher should preserve the exact source record and explain the relationship between that interpretation and the statutory provision under review. The notebook described here does not ingest these real judicial sources into its fictional vault.

The scholarly references provide methodological background at different levels. Work on legal technology and analytics frames the changing tasks of legal practitioners; research on retrieval-augmented generation explains why external evidence can be integrated into language-model workflows; work on reasoning-and-action patterns motivates an iterative agent; and scholarship on hallucination and the NIST AI Risk Management Framework informs risk identification and governance. These sources do not certify the proposed architecture, prove that an agent will perform well, or substitute for the experiment's own evaluation. They are included so that readers can distinguish the design's intellectual context from empirical findings that have yet to be produced.

For reproducibility, citations in any later memorandum should point to stable synthetic document identifiers and, where feasible, to a proposition or passage within the record. A legal citation in a real research product would require the corresponding official source, pinpoint reference, version information, and verification of authority. The two citation practices should never be conflated. The synthetic system can test whether an agent cites the records it actually inspected; it cannot establish that a citation to a fictional record would be adequate in professional practice. This distinction should be retained in teaching slides, screenshots, exported tables, and any demonstration derived from the notebook.

The bibliography is therefore both a research starting point and a boundary statement. Official statutes and portals belong to the real-world verification framework; fictional records belong to the experiment; and scholarly publications support the discussion of methods and risks. A later implementation should document which materials are used for background, which are encoded as synthetic examples, and which are retrieved during a run. If additional references are added, they should be identified by source type and relevance rather than merged into an undifferentiated list. This separation allows readers to audit the provenance of legal propositions and prevents the visual presence of a real statute in the references from lending false authenticity to an invented corpus item.

Readers should use the list as a verification map rather than as a closed canon. The selected sources illustrate where to begin for constitutional, statutory, administrative, judicial, technological, and governance questions, but the precise materials needed will depend on the facts and date of a real inquiry. Researchers should record exclusions as well as inclusions when a source appears relevant but is not authoritative for the issue. That practice makes the research path easier to review and discourages a model from treating every retrieved citation as equally persuasive.

Methodological note

SECTION OVERVIEW

This document is a proposed teaching and experiment design, not a report of completed results or a statement of current Mexican law. The shared Colab notebook is linked below; its access permissions, contents, and execution status should be independently verified before treating it as a completed experimental release.

This is an original pedagogical design specification, not a statement of current Mexican law or a report of completed experimental results. All quantities, performance criteria, legal examples, and architecture choices are proposals whose implementation and execution status should be checked in the linked Colab notebook. The document's claims should be read at the level of research design. It proposes a set of components, constraints, and measurements that could be implemented; it does not report that a particular agent has been deployed, that a data set has already been generated, or that one method has outperformed another. Prospective language is intentional. References to a 350-record corpus, five graph layers, three consultations, and ten notebook cells identify a planned teaching artifact, not a completed empirical system. Before publication of experimental results, the authors should replace any future-oriented descriptions with a record of what was actually implemented, including deviations from the proposed design.

The proposed corpus is a bounded fictional environment. The 350-document target is meant to balance enough variety for cross-domain navigation with the ability to inspect records and relationships. The planned source categories, metadata fields, exceptions, distractors, temporal flags, and factual predicates are design choices that should be fixed before comparisons begin. If the generated corpus differs in size or composition, the final notebook should report those differences and preserve the actual manifest. No real legal text should be presented as generated synthetic authority, and every fictional record should retain a synthetic label through export and citation. The corpus should be reviewed for internal consistency, but such review establishes consistency within the exercise, not truth about Mexican law.

The proposed agent has a narrow role: it selects among permitted research actions and supplies a rationale for the selection. A separate executor validates action names and parameters, retrieves records, logs results, and enforces limits. This allocation is a design hypothesis intended to keep model behavior from silently becoming execution authority. Implementation should specify the actual action schema, invalid-action response, retry policy, rate limits, and stopping conditions. If any of those rules change during development, the change should be versioned. The model identifier, service configuration, prompt, output schema, and fallback procedure should also be recorded because each can affect observed navigation behavior.

The three-method comparison is likewise a proposed protocol. Semantic retrieval, deterministic traversal, and autonomous navigation should receive the same consultation set and corpus version. Where budgets or model calls differ, the difference should be declared and either controlled or analyzed as a separate condition. Each run should preserve a unique identifier, random seed where applicable, timestamp, system configuration, and complete research trace. The notebook should define in advance how duplicate records, missing sources, failed API requests, malformed outputs, and incomplete runs will be treated. Excluding an inconvenient run after inspecting its score would undermine the purpose of a controlled comparison.

The instructor key and scoring rubric are measurement instruments, not self-validating ground truth. Critical records, distractors, necessary cross-domain links, and unresolved predicates should be listed with a reason for inclusion. At least a sample of the key should receive independent review, and disagreements should be documented. Metrics should be computed from the saved trace and generated memorandum rather than from memory or informal impressions. When human assessors judge citation support or uncertainty calibration, the rubric should define the unit being rated and provide examples of supported claims, qualified inferences, and unsupported assertions. If results later fail to support the motivating hypothesis, the report should preserve that outcome instead of rewriting the hypothesis after the fact.

Reproducibility requires more than a fixed seed. A complete teaching release should include the notebook, a readable description of the corpus schema, version information, configuration defaults, setup instructions, expected outputs, and a procedure for resetting state between methods. Secrets must remain outside the notebook and its outputs. The fallback mode should be able to demonstrate deterministic portions of the workflow without claiming that a language model made those decisions. Saved artifacts should make it possible to reconstruct the pathway from consultation through final metrics while excluding credentials and any unrelated personal data. A clean execution from the first cell to the last should be tested before the notebook link is circulated.

The design is intentionally limited in what it asks students to infer. It can support discussion of retrieval, evidence coverage, provenance, adaptive action selection, resource budgets, and auditability within a known synthetic environment. It cannot establish the legal correctness of a real memorandum, predict the behavior of an autonomous research system in every domain, or demonstrate that a model possesses professional judgment. Any transfer from this exercise to current Mexican legal practice requires separate validation against up-to-date official sources and qualified review. The experiment should be presented as a teaching tool for how to structure research decisions, not as a substitute for research responsibility.

The shared Colab artifact is linked below. The presence of a link does not establish that the notebook has been independently reviewed, runs successfully, or contains no credentials, private data, or misleading legal claims. Before treating it as a completed release, verify that its sharing permissions suit the intended audience, inspect its contents, run it from a fresh state, and confirm that it matches the corpus, methods, and metrics described in this monograph. Any later notebook revision that materially changes those elements should be accompanied by a version note so the written design and executable artifact remain synchronized. The final review should compare the notebook against this specification item by item. In particular, verify the number and purpose of cells, the synthetic labels, the action budget, the available fallback, the recorded evaluation measures, and the report's limitations. If a feature described here is omitted, deferred, or replaced, the document should be updated to state that clearly rather than leaving an outdated promise in place. The notebook's first shared version should include a small demonstration run and an explanation of how readers can reset the state, reproduce the output, and report a problem without sharing secrets. **Colab notebook:** Open the shared notebook on Google Drive.

Implementation protocol and release record

Before a notebook link is published, the written design and the executable artifact should be reconciled through a release checklist. The notebook should have a stable revision identifier and a clean run record, while this document should identify the same corpus schema, method definitions, action budgets, scoring rules, and limitations. Any divergence should be recorded in a short change note. Readers should be able to determine whether a described result is a proposal, an observed demonstration, or a measured experimental finding. This distinction is essential because a polished design document can otherwise make planned controls appear to have been implemented and tested.

The publication review should include a reproducibility check by someone other than the author. That reviewer should open the shared copy, confirm the permissions, run the notebook from a fresh state, inspect the saved trace and outputs, and verify that no secret, private consultation, or misleading legal claim appears in the shared material. The URL should resolve to the exact version referenced in the monograph, not to an evolving working copy whose contents can change without notice. If later revisions alter the corpus, models, prompt, or metrics, retain the prior version or a manifest sufficient to identify what changed. A URL locates an artifact; its version record makes the experiment interpretable.

Claims in the paper should be categorized before release. Design intentions use prospective language; demonstrations report only what the run shows; empirical claims state sample sizes, conditions, and uncertainty; legal statements about real authority require current authoritative verification and qualified review. This claim inventory can be applied sentence by sentence to ensure that synthetic examples are not mistaken for legal rules and that experimental proposals are not described as findings. The methodological note should be updated whenever a promised feature is deferred, replaced, or found infeasible. A transparent limitation is more defensible than an unqualified claim that the notebook is complete.

Finally, release governance should specify a contact and process for reporting an error, requesting correction, or identifying a safety concern. Reports should be logged with an issue identifier, affected version, severity, decision, and resolution. If a defect affects citations, source status, or a published result, the team should assess whether a correction notice or withdrawal is necessary. These procedures are appropriate even for a teaching artifact because the document may be copied, shared, or used as a template for higher-risk work. The link belongs in the artifact's version-control and accountability record.

REAL-WORLD IMPLEMENTATION CHECKLIST

- Publish a release manifest tying the paper, notebook revision, corpus schema, model configuration, prompts, and scoring protocol together.
- Have an independent reviewer reproduce the clean run, inspect sharing permissions, and confirm that outputs expose no secrets or private facts.

- Label every claim as a proposal, demonstration, empirical result, or verified legal statement and apply the appropriate evidentiary standard.
- Maintain versioned issue reports for citation defects, safety concerns, corrections, and decisions to amend or withdraw an artifact.
- Verify that the shared URL resolves to the reviewed release, and archive enough version information to reconstruct the linked experiment.